

ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 2 — Daily Curriculum: Months 1-4

112 daily cards with exact resources, labs, evidence, retrieval and completion criteria

Edition: Volume 2
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Month 01 — Computer systems, IT foundations, operating systems and command line

Days D001-D028. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Baseline diagnostic and binary number systems D2: Computer organization and the fetch-decode-execute cycle D3: Operating-system architecture and abstraction D4: Virtualization and lab safety D5: Command-line navigation and file operations D6: Users, groups, ownership and permissions D7: Home lab baseline project and evidence workflow
Week 2	D1: Weekly retrieval, error-log cleanup and catch-up D2: Filesystem hierarchy, inodes and links D3: Processes, threads, scheduling and signals D4: Services, units, timers and system logs D5: Client-server architecture and sockets D6: HTTP requests, responses and status codes D7: Databases and SQL fundamentals
Week 3	D1: Git, branching and merge conflicts D2: Weekly review and mini practical examination D3: Memory, stack, heap and addresses D4: Binary logic and CPU instruction foundations D5: Storage, filesystems, journaling and backups D6: HTTP sessions, cookies and authentication basics D7: Troubleshooting methodology and technical journaling
Week 4	D1: Month 1 project completion and portfolio publication D2: Month 1 practical assessment and milestone review D3: APIs, JSON and structured data D4: Identity, sessions and access control foundations D5: Virtual machine networking troubleshooting D6: CS50-style problem solving and decomposition D7: Spaced revision, recovery and Month 2 preview

> **Gate:** score the Month 1 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 01 · Week 1 · daily cards

D001 · M01W1D1 · Baseline diagnostic and binary number systems

Phase: Academic month · **Optional date:** 2026-10-27 · **Domain resources:** F01 / F06

1. Day / sequence	Global Day 1 of 365; calendar-free M01W1D1; optional date 2026-10-27.
2. Exact topic	Baseline diagnostic and binary number systems
3. Exact subtopics	Decimal, binary and hexadecimal; bits and bytes; signed/unsigned integers; character encodings
4. Learning objective	Convert values between bases and explain why computers use binary
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 0, Scratch and binary representation lesson; do not skip the problem set. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a 20-question self-diagnostic on number systems, programming and networking; create a baseline score file. **2.** Standard (~2 h): Run a 20-question self-diagnostic on number systems, programming and networking; create a baseline score file. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a 20-question self-diagnostic on number systems, programming and networking; create a baseline score file. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	diagnostic.csv and a one-page learning contract; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is 255 the largest unsigned decimal value in one byte, and where is this relevant to IPv4? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score logged, errors classified, and learning contract committed to the journal Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 0, Scratch and binary representation lesson; do not skip the problem set"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D002 · M01W1D2 · Computer organization and the fetch-decode-execute cycle

Phase: Academic month · **Optional date:** 2026-10-28 · **Domain resources:** F07 / F06

1. Day / sequence	Global Day 2 of 365; calendar-free M01W1D2; optional date 2026-10-28.
2. Exact topic	Computer organization and the fetch-decode-execute cycle
3. Exact subtopics	CPU, control unit, ALU, registers, cache, RAM, storage, buses, clock and instruction cycle
4. Learning objective	Trace the path of one instruction and distinguish volatile from persistent storage
5. Primary lecture / reading + direct URL	Nand2Tetris / The Elements of Computing Systems `https://www.nand2tetris.org/` **Type:** Free course and software · **Access:** Free software and course materials; book optional paid
6. Exact section / lesson locator	Nand2Tetris Part I, Project 1 overview and Project 2 arithmetic introduction. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a labelled block diagram of a simple computer from input to output; annotate each component's security relevance. **2.** Standard (~2 h): Build a labelled block diagram of a simple computer from input to output; annotate each component's security relevance. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a labelled block diagram of a simple computer from input to output; annotate each component's security relevance. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	architecture.svg and a 150-word explanation; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would an attacker gain from direct memory access, and which layer prevents it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram has all required components and the explanation uses correct terminology Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nand2Tetris Part I, Project 1 overview and Project 2 arithmetic introduction"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Operating Systems: Three Easy Pieces (OSTEP); write a five-line correction; retest next morning with a new 10-minute problem.

D003 · M01W1D3 · Operating-system architecture and abstraction

Phase: Academic month · **Optional date:** 2026-10-29 · **Domain resources:** F06 / F01

1. Day / sequence	Global Day 3 of 365; calendar-free M01W1D3; optional date 2026-10-29.
2. Exact topic	Operating-system architecture and abstraction
3. Exact subtopics	Kernel mode versus user mode, system calls, device drivers, virtual memory, process isolation, multitasking
4. Learning objective	Explain how an OS provides abstractions and enforces isolation
5. Primary lecture / reading + direct URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/` **Type:** Free textbook · **Access:** Free online chapters; paid print optional
6. Exact section / lesson locator	OSTEP Chapter 2 Introduction and Chapter 4 Processes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Observe process creation on Linux or Windows, record PID/PPID and map five OS abstractions in a table. **2.** Standard (~2 h): Observe process creation on Linux or Windows, record PID/PPID and map five OS abstractions in a table. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Observe process creation on Linux or Windows, record PID/PPID and map five OS abstractions in a table. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	process-observation.md with real command output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can one crashed application usually be terminated without rebooting the OS? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Observed evidence matches the explanation and all five abstractions are named correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OSTEP Chapter 2 Introduction and Chapter 4 Processes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x 2026 OpenCourseWare; write a five-line correction; retest next morning with a new 10-minute problem.

D004 · M01W1D4 · Virtualization and lab safety

Phase: Academic month · **Optional date:** 2026-10-30 · **Domain resources:** F12 / F13

1. Day / sequence	Global Day 4 of 365; calendar-free M01W1D4; optional date 2026-10-30.
2. Exact topic	Virtualization and lab safety
3. Exact subtopics	Hypervisors, virtual machines, snapshots, NAT/host-only/internal networking, resource allocation, attack surface
4. Learning objective	Create an isolated Ubuntu VM with a clean snapshot and a documented network mode
5. Primary lecture / reading + direct URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	VirtualBox User Manual, Chapters 1-3, 5-6 and 8 on setup, storage, networking and snapshots. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Ubuntu Desktop Guide `https://help.ubuntu.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Install VirtualBox and Ubuntu LTS in a VM; create NAT plus host-only/internal adapters; take a clean snapshot. **2.** Standard (~2 h): Install VirtualBox and Ubuntu LTS in a VM; create NAT plus host-only/internal adapters; take a clean snapshot. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Install VirtualBox and Ubuntu LTS in a VM; create NAT plus host-only/internal adapters; take a clean snapshot. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	VM screenshots, network settings export and lab-setup.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must vulnerable services not be bridged to a public or shared network? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	VM boots, has internet only through NAT where intended, and a restore point exists Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "VirtualBox User Manual, Chapters 1-3, 5-6 and 8 on setup, storage, networking and snapshots"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D005 · M01W1D5 · Command-line navigation and file operations

Phase: Academic month · **Optional date:** 2026-10-31 · **Domain resources:** L01 / F08

1. Day / sequence	Global Day 5 of 365; calendar-free M01W1D5; optional date 2026-10-31.
2. Exact topic	Command-line navigation and file operations
3. Exact subtopics	Shell prompt, absolute/relative paths, globbing, ls/cd/pwd/mkdir/touch/cp/mv/rm, wildcards and quoting
4. Learning objective	Navigate and manipulate files safely without a graphical file manager
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Command Line and Filesystem Hierarchy sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 25 navigation and file operations in the VM using only the terminal; include one deliberate quoting error and its fix. **2.** Standard (~2 h): Complete 25 navigation and file operations in the VM using only the terminal; include one deliberate quoting error and its fix. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 25 navigation and file operations in the VM using only the terminal; include one deliberate quoting error and its fix. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	commands.log and errors.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between rm -r and rm -rf, and why does quoting matter? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All required operations succeed and the error is documented with a safe correction Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Command Line and Filesystem Hierarchy sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D006 · M01W1D6 · Users, groups, ownership and permissions

Phase: Academic month · **Optional date:** 2026-11-01 · **Domain resources:** L01 / F06

1. Day / sequence	Global Day 6 of 365; calendar-free M01W1D6; optional date 2026-11-01.
2. Exact topic	Users, groups, ownership and permissions
3. Exact subtopics	Discretionary access control, rwx bits, umask, sudo, principle of least privilege, ownership
4. Learning objective	Create users and groups and justify a least-privilege file policy
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Users and Groups and Permissions sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create two users and one group, set directory permissions for a shared lab folder, and verify access as each user. **2.** Standard (~2 h): Create two users and one group, set directory permissions for a shared lab folder, and verify access as each user. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create two users and one group, set directory permissions for a shared lab folder, and verify access as each user. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	permission-matrix.csv and terminal transcript; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does execute permission on a directory mean traversal rather than program execution? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each test user can perform only the intended operations Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Users and Groups and Permissions sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D007 · M01W1D7 · Home lab baseline project and evidence workflow

Phase: Academic month · **Optional date:** 2026-11-02 · **Domain resources:** F09 / F12 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 7 of 365; calendar-free M01W1D7; optional date 2026-11-02.
2. Exact topic	Home lab baseline project and evidence workflow
3. Exact subtopics	Asset inventory, baseline configuration, diagram, evidence integrity, README and Git commit
4. Learning objective	Produce the first reproducible home-lab deliverable
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git, Chapters 1-2: getting started and Git basics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a repository for the lab, add topology, VM details, setup steps, hashes of exported settings, and a checklist; commit in small units — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a repository for the lab, add topology, VM details, setup steps, hashes of exported settings, and a checklist; commit in small units — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a repository for the lab, add topology, VM details, setup steps, hashes of exported settings, and a checklist; commit in small units — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-01 repository with README, diagram, evidence and clean commit history; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which three facts must another person know to reproduce your VM network? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository has no secrets, builds from clean instructions, and contains a tagged v0.1 release Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git, Chapters 1-2: getting started and Git basics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

Month 01 · Week 2 · daily cards

D008 · M01W2D1 · Weekly retrieval, error-log cleanup and catch-up

Phase: Academic month · **Optional date:** 2026-11-03 · **Domain resources:** F01 / F06

1. Day / sequence	Global Day 8 of 365; calendar-free M01W2D1; optional date 2026-11-03.
2. Exact topic	Weekly retrieval, error-log cleanup and catch-up
3. Exact subtopics	Number systems, OS abstractions, VM isolation, CLI, users and permissions; spaced repetition
4. Learning objective	Repair gaps found by the diagnostic and prepare for Week 2
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 0 notes and Week 1 C introduction; use the official notes page. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Do 15 closed-book retrieval questions, re-run three failed commands, and rewrite two explanations from memory. **2.** Standard (~2 h): Do 15 closed-book retrieval questions, re-run three failed commands, and rewrite two explanations from memory. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Do 15 closed-book retrieval questions, re-run three failed commands, and rewrite two explanations from memory. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	weekly-review-M1W1.md with question scores and corrected answers; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Explain the chain from CPU instruction to OS process to user file access without notes Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70 percent and all previous unsafe lab mistakes have corrections Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 0 notes and Week 1 C introduction; use the official notes page"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D009 · M01W2D2 · Filesystem hierarchy, inodes and links

Phase: Academic month · **Optional date:** 2026-11-04 · **Domain resources:** L01 / L04

1. Day / sequence	Global Day 9 of 365; calendar-free M01W2D2; optional date 2026-11-04.
2. Exact topic	Filesystem hierarchy, inodes and links
3. Exact subtopics	Root filesystem, mount points, inodes, hard links, symbolic links, file metadata, ext4 basics
4. Learning objective	Explain how paths resolve to storage and why files persist after a name is removed
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Filesystem Hierarchy and Filesystem sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use stat, df, mount, ln and ln -s on benign files; delete a hard link and observe link counts. **2.** Standard (~2 h): Use stat, df, mount, ln and ln -s on benign files; delete a hard link and observe link counts. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use stat, df, mount, ln and ln -s on benign files; delete a hard link and observe link counts. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	filesystem-notes.md with command output and diagrams; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a symbolic link break while a hard link count changes? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All observations match the predicted link and inode behaviour Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Filesystem Hierarchy and Filesystem sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D010 · M01W2D3 · Processes, threads, scheduling and signals

Phase: Academic month · **Optional date:** 2026-11-05 · **Domain resources:** F06 / L01

1. Day / sequence	Global Day 10 of 365; calendar-free M01W2D3; optional date 2026-11-05.
2. Exact topic	Processes, threads, scheduling and signals
3. Exact subtopics	Process lifecycle, PIDs, parent/child relationships, threads, context switching, SIGTERM versus SIGKILL
4. Learning objective	Observe process state and explain how the CPU shares time
5. Primary lecture / reading + direct URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/` **Type:** Free textbook · **Access:** Free online chapters; paid print optional
6. Exact section / lesson locator	OSTEP Chapters 4-7: Processes, Process API, Direct Execution and CPU Scheduling. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Start a benign background process, inspect it with ps/top/pstree, send TERM and then repeat with KILL; record state changes. **2.** Standard (~2 h): Start a benign background process, inspect it with ps/top/pstree, send TERM and then repeat with KILL; record state changes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Start a benign background process, inspect it with ps/top/pstree, send TERM and then repeat with KILL; record state changes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	process-lab.md with timestamps and signal results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is SIGKILL a last resort and what evidence might be lost? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Process tree, state transitions and signal results are correctly explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OSTEP Chapters 4-7: Processes, Process API, Direct Execution and CPU Scheduling"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x 2026 OpenCourseWare; write a five-line correction; retest next morning with a new 10-minute problem.

D011 · M01W2D4 · Services, units, timers and system logs

Phase: Academic month · **Optional date:** 2026-11-06 · **Domain resources:** L05 / L01

1. Day / sequence	Global Day 11 of 365; calendar-free M01W2D4; optional date 2026-11-06.
2. Exact topic	Services, units, timers and system logs
3. Exact subtopics	systemd units, systemctl, journalctl, dependencies, restart policy and log levels
4. Learning objective	Create, inspect and remove a benign service and timer safely
5. Primary lecture / reading + direct URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	systemd.unit manual, unit structure and dependency sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a harmless timer that writes the time to a local file, inspect journal logs, then disable and remove the unit. **2.** Standard (~2 h): Create a harmless timer that writes the time to a local file, inspect journal logs, then disable and remove the unit. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a harmless timer that writes the time to a local file, inspect journal logs, then disable and remove the unit. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	unit-file.txt, journal excerpts and cleanup proof; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between a service failing and a service being inactive? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Service ran, was observed, and no residual unit remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "systemd.unit manual, unit structure and dependency sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux man-pages project; write a five-line correction; retest next morning with a new 10-minute problem.

D012 · M01W2D5 · Client-server architecture and sockets

Phase: Academic month · **Optional date:** 2026-11-07 · **Domain resources:** F10 / N14

1. Day / sequence	Global Day 12 of 365; calendar-free M01W2D5; optional date 2026-11-07.
2. Exact topic	Client-server architecture and sockets
3. Exact subtopics	Clients, servers, listeners, localhost, loopback, IP address, port, socket, bind/listen/accept/connect
4. Learning objective	Map a local client-server interaction to ports and sockets
5. Primary lecture / reading + direct URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview` **Type:** Official web documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP Overview, HTTP basics and client-server communication sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	IANA Service Name and Transport Protocol Port Number Registry `https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Start a local Python HTTP server on loopback, connect with curl/browser, and identify local and remote addresses and ports. **2.** Standard (~2 h): Start a local Python HTTP server on loopback, connect with curl/browser, and identify local and remote addresses and ports. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Start a local Python HTTP server on loopback, connect with curl/browser, and identify local and remote addresses and ports. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	client-server.md and ss/netstat output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can two different services use port 80 on different hosts but not the same IP simultaneously? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Loopback service worked and all socket fields are labelled correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP Overview, HTTP basics and client-server communication sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D013 · M01W2D6 · HTTP requests, responses and status codes

Phase: Academic month · **Optional date:** 2026-11-08 · **Domain resources:** F10 / W04

1. Day / sequence	Global Day 13 of 365; calendar-free M01W2D6; optional date 2026-11-08.
2. Exact topic	HTTP requests, responses and status codes
3. Exact subtopics	Methods, URL, request headers, response headers, status code families, content types, redirects
4. Learning objective	Read a raw HTTP exchange and explain each field's purpose
5. Primary lecture / reading + direct URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview` **Type:** Official web documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP Overview and HTTP methods/status sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MDN HTTP documentation `https://developer.mozilla.org/en-US/docs/Web/HTTP`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use curl -v or browser DevTools against a local test server; save one request/response pair and annotate headers. **2.** Standard (~2 h): Use curl -v or browser DevTools against a local test server; save one request/response pair and annotate headers. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use curl -v or browser DevTools against a local test server; save one request/response pair and annotate headers. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	http-annotated.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between 301, 302, 401 and 403 in practical terms? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All headers and status codes in the capture are correctly interpreted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP Overview and HTTP methods/status sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D014 · M01W2D7 · Databases and SQL fundamentals

Phase: Academic month · **Optional date:** 2026-11-09 · **Domain resources:** F11 / P07 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 14 of 365; calendar-free M01W2D7; optional date 2026-11-09.
2. Exact topic	Databases and SQL fundamentals
3. Exact subtopics	Tables, rows, columns, primary keys, SELECT, INSERT, WHERE, ORDER BY and LIMIT
4. Learning objective	Query and insert synthetic security events in SQLite
5. Primary lecture / reading + direct URL	SQLBolt `https://sqlbolt.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	SQLBolt Lessons 1-5: SELECT queries, filtering and sorting. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SQLite documentation `https://www.sqlite.org/docs.html`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a small auth_events table, insert ten synthetic rows and write five queries with expected outputs — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a small auth_events table, insert ten synthetic rows and write five queries with expected outputs — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a small auth_events table, insert ten synthetic rows and write five queries with expected outputs — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	schema.sql, queries.sql and results.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why should production code use parameterized queries rather than string concatenation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All five queries run and results are verified against expected rows Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLBolt Lessons 1-5: SELECT queries, filtering and sorting"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

Month 01 · Week 3 · daily cards

D015 · M01W3D1 · Git, branching and merge conflicts

Phase: Academic month · **Optional date:** 2026-11-10 · **Domain resources:** F09 / F01

1. Day / sequence	Global Day 15 of 365; calendar-free M01W3D1; optional date 2026-11-10.
2. Exact topic	Git, branching and merge conflicts
3. Exact subtopics	Repository initialization, staging, commits, branches, diffs, merge, conflict markers and .gitignore
4. Learning objective	Use version control to recover from a deliberate mistake
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git, Chapters 2-3: Git basics and branching. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a repository, make two branches that modify the same line, resolve the conflict, and add a .gitignore for lab secrets. **2.** Standard (~2 h): Create a repository, make two branches that modify the same line, resolve the conflict, and add a .gitignore for lab secrets. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a repository, make two branches that modify the same line, resolve the conflict, and add a .gitignore for lab secrets. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	git-log.txt and resolved conflict screenshot; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between git reset, revert and checkout in your own workflow? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Clean history, conflict resolved, and secret placeholders ignored Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git, Chapters 2-3: Git basics and branching"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D016 · M01W3D2 · Weekly review and mini practical examination

Phase: Academic month · **Optional date:** 2026-11-11 · **Domain resources:** F11 / F09

1. Day / sequence	Global Day 16 of 365; calendar-free M01W3D2; optional date 2026-11-11.
2. Exact topic	Weekly review and mini practical examination
3. Exact subtopics	Filesystems, processes, services, sockets, HTTP, SQL and Git; active recall and troubleshooting
4. Learning objective	Demonstrate Month 1 Week 2 skills under a timed scenario
5. Primary lecture / reading + direct URL	SQLBolt `https://sqlbolt.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	SQLBolt Lessons 6-10: joins, aggregation and NULL values. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 60-minute practical: inspect a process, start a loopback service, query a SQLite table, and commit evidence. **2.** Standard (~2 h): Complete a 60-minute practical: inspect a process, start a loopback service, query a SQLite table, and commit evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 60-minute practical: inspect a process, start a loopback service, query a SQLite table, and commit evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	timed-practical-M1W2.md with all command output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Given a broken service, which three logs or commands would you check first and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least 70 percent on the rubric and all evidence is reproducible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLBolt Lessons 6-10: joins, aggregation and NULL values"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D017 · M01W3D3 · Memory, stack, heap and addresses

Phase: Academic month · **Optional date:** 2026-11-12 · **Domain resources:** F01 / F06

1. Day / sequence	Global Day 17 of 365; calendar-free M01W3D3; optional date 2026-11-12.
2. Exact topic	Memory, stack, heap and addresses
3. Exact subtopics	Virtual memory, address space, stack frames, heap allocation, pointers, buffer boundaries
4. Learning objective	Describe how data and control information are laid out in memory
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 4 Memory lecture and notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a safe C/Python memory visualization exercise; draw stack and heap regions for a simple function call. **2.** Standard (~2 h): Run a safe C/Python memory visualization exercise; draw stack and heap regions for a simple function call. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a safe C/Python memory visualization exercise; draw stack and heap regions for a simple function call. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	memory-diagram.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the security consequence of writing beyond an array boundary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram correctly locates code, globals, heap, stack and function frames Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 4 Memory lecture and notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D018 · M01W3D4 · Binary logic and CPU instruction foundations

Phase: Academic month · **Optional date:** 2026-11-13 · **Domain resources:** F07 / F02

1. Day / sequence	Global Day 18 of 365; calendar-free M01W3D4; optional date 2026-11-13.
2. Exact topic	Binary logic and CPU instruction foundations
3. Exact subtopics	Logic gates, truth tables, ALU operations, machine language, registers and program counter
4. Learning objective	Connect Boolean logic to a CPU instruction and a security decision
5. Primary lecture / reading + direct URL	Nand2Tetris / The Elements of Computing Systems `https://www.nand2tetris.org/` **Type:** Free course and software · **Access:** Free software and course materials; book optional paid
6. Exact section / lesson locator	Nand2Tetris Part I Projects 1-3: logic gates, arithmetic and sequential logic. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50x Lectures 2026 (official YouTube playlist) `https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete one software simulator project or, if unavailable, write truth tables and explain a half-adder and flip-flop. **2.** Standard (~2 h): Complete one software simulator project or, if unavailable, write truth tables and explain a half-adder and flip-flop. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete one software simulator project or, if unavailable, write truth tables and explain a half-adder and flip-flop. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	logic-project files or truth-tables.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How does a branch instruction change control flow at the hardware level? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Project passes local checks or all truth tables are correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nand2Tetris Part I Projects 1-3: logic gates, arithmetic and sequential logic"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Operating Systems: Three Easy Pieces (OSTEP); write a five-line correction; retest next morning with a new 10-minute problem.

D019 · M01W3D5 · Storage, filesystems, journaling and backups

Phase: Academic month · **Optional date:** 2026-11-14 · **Domain resources:** F12 / C27

1. Day / sequence	Global Day 19 of 365; calendar-free M01W3D5; optional date 2026-11-14.
2. Exact topic	Storage, filesystems, journaling and backups
3. Exact subtopics	Block devices, partitions, filesystem metadata, journaling, snapshots, 3-2-1 backup concept, restore testing
4. Learning objective	Explain why an untested backup is not a valid recovery control
5. Primary lecture / reading + direct URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	VirtualBox User Manual virtual storage and snapshots chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-34 Revision 1 `https://csrc.nist.gov/pubs/sp/800/34/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a benign file set in the VM, snapshot or copy it, corrupt/delete one file, and restore; record hashes before and after. **2.** Standard (~2 h): Create a benign file set in the VM, snapshot or copy it, corrupt/delete one file, and restore; record hashes before and after. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a benign file set in the VM, snapshot or copy it, corrupt/delete one file, and restore; record hashes before and after. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	backup-test.md with SHA-256 values; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between a VM snapshot, a file backup and a disaster-recovery plan? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Restore is proven by matching hashes and cleanup is documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "VirtualBox User Manual virtual storage and snapshots chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D020 · M01W3D6 · HTTP sessions, cookies and authentication basics

Phase: Academic month · **Optional date:** 2026-11-15 · **Domain resources:** F04 / W17

1. Day / sequence	Global Day 20 of 365; calendar-free M01W3D6; optional date 2026-11-15.
2. Exact topic	HTTP sessions, cookies and authentication basics
3. Exact subtopics	Statefulness, cookies, Set-Cookie attributes, sessions, authentication factors and password hashing concept
4. Learning objective	Distinguish authentication from authorization and explain safe password storage
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 0 Securing Accounts and Week 1 Securing Data notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Inspect a local app's cookie in DevTools; annotate Secure, HttpOnly, SameSite and explain password hashing versus encryption. **2.** Standard (~2 h): Inspect a local app's cookie in DevTools; annotate Secure, HttpOnly, SameSite and explain password hashing versus encryption. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Inspect a local app's cookie in DevTools; annotate Secure, HttpOnly, SameSite and explain password hashing versus encryption. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cookies-notes.md and a conceptual login-flow diagram; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is storing plaintext passwords worse than storing unsalted fast hashes? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All cookie attributes and identity terms are explained without confusing hashing with encryption Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 0 Securing Accounts and Week 1 Securing Data notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D021 · M01W3D7 · Troubleshooting methodology and technical journaling

Phase: Academic month · **Optional date:** 2026-11-16 · **Domain resources:** F01 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 21 of 365; calendar-free M01W3D7; optional date 2026-11-16.
2. Exact topic	Troubleshooting methodology and technical journaling
3. Exact subtopics	Define symptom, establish baseline, form hypotheses, change one variable, collect evidence, rollback and document
4. Learning objective	Diagnose a deliberately broken local service using a repeatable method
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 1 C problem-solving and debugging sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Break a local HTTP server configuration on purpose, diagnose it from logs, fix it, and write a postmortem — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Break a local HTTP server configuration on purpose, diagnose it from logs, fix it, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Break a local HTTP server configuration on purpose, diagnose it from logs, fix it, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	troubleshooting-postmortem.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which observation falsified each incorrect hypothesis? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Root cause is proven and rollback steps are included Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 1 C problem-solving and debugging sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

Month 01 · Week 4 · daily cards

D022 · M01W4D1 · Month 1 project completion and portfolio publication

Phase: Academic month · **Optional date:** 2026-11-17 · **Domain resources:** F09 / F12

1. Day / sequence	Global Day 22 of 365; calendar-free M01W4D1; optional date 2026-11-17.
2. Exact topic	Month 1 project completion and portfolio publication
3. Exact subtopics	Problem statement, architecture, validation, limitations, screenshots, ethical scope and README standards
4. Learning objective	Publish a sanitized, reproducible home-lab baseline report
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapter 2 and GitHub Docs basic README conventions via F09's official site navigation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Finalize PRJ-01, add verification tests, remove IP/MAC addresses or personal data, and tag v0.2. **2.** Standard (~2 h): Finalize PRJ-01, add verification tests, remove IP/MAC addresses or personal data, and tag v0.2. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Finalize PRJ-01, add verification tests, remove IP/MAC addresses or personal data, and tag v0.2. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published PRJ-01 repository and report PDF/Markdown; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would a reviewer check first to verify this lab actually works? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository passes the Month 1 rubric and contains no secrets or personal data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapter 2 and GitHub Docs basic README conventions via F09's official site navigation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D023 · M01W4D2 · Month 1 practical assessment and milestone review

Phase: Academic month · **Optional date:** 2026-11-18 · **Domain resources:** F01 / F04

1. Day / sequence	Global Day 23 of 365; calendar-free M01W4D2; optional date 2026-11-18.
2. Exact topic	Month 1 practical assessment and milestone review
3. Exact subtopics	Number systems, architecture, OS, virtualization, command line, HTTP, SQL, Git and troubleshooting
4. Learning objective	Demonstrate foundational competence through an independent 90-minute practical
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 1-4 notes for algorithmic reasoning and memory. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Without notes, configure a fresh lightweight VM, create users, run a local service, query a synthetic database, commit evidence and answer 20 oral questions. **2.** Standard (~2 h): Without notes, configure a fresh lightweight VM, create users, run a local service, query a synthetic database, commit evidence and answer 20 oral questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Without notes, configure a fresh lightweight VM, create users, run a local service, query a synthetic database, commit evidence and answer 20 oral questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	assessment-M1.pdf with command evidence and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Explain how a request travels from browser to local server and where isolation occurs Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 with no safety violation and practical section at least 18/25 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 1-4 notes for algorithmic reasoning and memory"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D024 · M01W4D3 · APIs, JSON and structured data

Phase: Academic month · **Optional date:** 2026-11-19 · **Domain resources:** F10 / P06

1. Day / sequence	Global Day 24 of 365; calendar-free M01W4D3; optional date 2026-11-19.
2. Exact topic	APIs, JSON and structured data
3. Exact subtopics	REST resources, HTTP verbs, status codes, JSON objects/arrays, serialization and schema validation
4. Learning objective	Read and design a simple API request/response pair with explicit validation
5. Primary lecture / reading + direct URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview` **Type:** Official web documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP Overview and HTTP methods/status sections; F08 Python JSON tutorial section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Requests: HTTP for Humans `https://requests.readthedocs.io/en/latest/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use a local mock server or Python requests against localhost; send valid, malformed and missing-field JSON; record status codes and errors. **2.** Standard (~2 h): Use a local mock server or Python requests against localhost; send valid, malformed and missing-field JSON; record status codes and errors. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use a local mock server or Python requests against localhost; send valid, malformed and missing-field JSON; record status codes and errors. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	api-notes.md and sample request/response files; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must a client validate server data even when it sent valid JSON? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All three cases are documented and status codes are interpreted correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP Overview and HTTP methods/status sections; F08 Python JSON tutorial section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D025 · M01W4D4 · Identity, sessions and access control foundations

Phase: Academic month · **Optional date:** 2026-11-20 · **Domain resources:** F04 / W17

1. Day / sequence	Global Day 25 of 365; calendar-free M01W4D4; optional date 2026-11-20.
2. Exact topic	Identity, sessions and access control foundations
3. Exact subtopics	Identification, authentication, authorization, accountability, session identifiers, logout and least privilege
4. Learning objective	Map a login workflow to identity controls and list three session risks
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 0 Securing Accounts notes and Week 3 Securing Software notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a local login sequence, mark where identity is established and checked, and describe session fixation and timeout risks. **2.** Standard (~2 h): Draw a local login sequence, mark where identity is established and checked, and describe session fixation and timeout risks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a local login sequence, mark where identity is established and checked, and describe session fixation and timeout risks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	identity-flow.svg and risk notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a long-lived session identifier dangerous after a password change? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram separates authentication from authorization and identifies concrete controls Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 0 Securing Accounts notes and Week 3 Securing Software notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D026 · M01W4D5 · Virtual machine networking troubleshooting

Phase: Academic month · **Optional date:** 2026-11-21 · **Domain resources:** F12 / N08

1. Day / sequence	Global Day 26 of 365; calendar-free M01W4D5; optional date 2026-11-21.
2. Exact topic	Virtual machine networking troubleshooting
3. Exact subtopics	Virtual NIC types, NAT forwarding, host-only addressing, internal network, promiscuous mode, DNS and DHCP inside VMs
4. Learning objective	Diagnose VM connectivity without weakening host security
5. Primary lecture / reading + direct URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	VirtualBox User Manual Chapter 6 Virtual Networking and Chapter 8 VBoxManage/network troubleshooting. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create three harmless VM network faults, diagnose each from guest and host, then restore the clean snapshot and document commands. **2.** Standard (~2 h): Create three harmless VM network faults, diagnose each from guest and host, then restore the clean snapshot and document commands. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create three harmless VM network faults, diagnose each from guest and host, then restore the clean snapshot and document commands. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vm-network-troubleshooting.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is promiscuous mode risky on a shared network and when might it be needed? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each fault has evidence-based diagnosis and the clean snapshot is restored Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "VirtualBox User Manual Chapter 6 Virtual Networking and Chapter 8 VBoxManage/network troubleshooting"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D027 · M01W4D6 · CS50-style problem solving and decomposition

Phase: Academic month · **Optional date:** 2026-11-22 · **Domain resources:** F01 / F08

1. Day / sequence	Global Day 27 of 365; calendar-free M01W4D6; optional date 2026-11-22.
2. Exact topic	CS50-style problem solving and decomposition
3. Exact subtopics	Abstraction, algorithms, pseudocode, inputs/outputs, edge cases, test cases and debugging
4. Learning objective	Solve a small unfamiliar programming problem by decomposing it before coding
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 1 C lecture and problem-solving notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write pseudocode and a Python implementation for a log-line classifier; test empty input, large input, malformed input and unexpected characters. **2.** Standard (~2 h): Write pseudocode and a Python implementation for a log-line classifier; test empty input, large input, malformed input and unexpected characters. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write pseudocode and a Python implementation for a log-line classifier; test empty input, large input, malformed input and unexpected characters. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	classifier.py and test-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which edge case caused a failure and how did you change the algorithm? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Code passes all planned tests and the write-up explains the decomposition Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 1 C lecture and problem-solving notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D028 · M01W4D7 · Spaced revision, recovery and Month 2 preview

Phase: Academic month · **Optional date:** 2026-11-23 · **Domain resources:** N16 / F06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 28 of 365; calendar-free M01W4D7; optional date 2026-11-23.
2. Exact topic	Spaced revision, recovery and Month 2 preview
3. Exact subtopics	Error log, command notebook, flashcards, weak-topic repair, subnetting readiness and binary practice
4. Learning objective	Consolidate Month 1 and prepare binary arithmetic for IP addressing
5. Primary lecture / reading + direct URL	Subnetting Practice `https://subnettingpractice.com/` **Type:** Interactive practice · **Access:** Free tier; optional ads/premium features may appear
6. Exact section / lesson locator	Subnetting Practice, introduction and binary/IP basics; do only five diagnostic questions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Review all failed retrieval questions, convert 20 decimal values to binary, write five Anki cards, and plan the next week's lab — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Review all failed retrieval questions, convert 20 decimal values to binary, write five Anki cards, and plan the next week's lab — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Review all failed retrieval questions, convert 20 decimal values to binary, write five Anki cards, and plan the next week's lab — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	revision-M1.md and flashcards.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which Month 1 concept is still weakest, and what exact exercise would repair it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All weak topics have a scheduled repair action and at least 18/20 binary conversions are correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Subnetting Practice, introduction and binary/IP basics; do only five diagnostic questions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Practical Networking Subnetting Mastery; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 — Networking, subnetting, packet analysis and troubleshooting

Days D029-D056. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: OSI seven-layer model and TCP/IP four-layer model D2: Ethernet frames, MAC addresses and switching D3: ARP, ICMP and local address resolution D4: VLANs, trunks and broadcast domains D5: IPv4 header, addresses and private ranges D6: CIDR, subnet masks and network/broadcast/host calculations D7: VLSM, route aggregation and daily subnetting fluency
Week 2	D1: Weekly packet and subnetting review D2: Routing, default gateways and static routes D3: NAT, PAT and public versus private addressing D4: Dynamic routing and routing security concepts D5: TCP handshake, sequence numbers and reliability D6: UDP, ports and common services D7: DNS resolution and record types
Week 3	D1: DHCP, DHCPv6 and network configuration lifecycle D2: HTTP, HTTPS and TLS at the network layer D3: Weekly network services practical and retrieval D4: Firewalls, proxies, VPNs and segmentation architectures D5: Wireshark display filters and protocol dissection D6: Nmap host discovery and port scanning in an isolated lab D7: TCP retransmissions, MTU and fragmentation analysis
Week 4	D1: Network troubleshooting methodology and tool selection D2: Network-security project: topology and traffic report D3: Monthly network practical and daily subnetting test D4: Monthly review and gap repairs D5: IPv6 addressing and protocol differences D6: Wireless LANs, network monitoring and SPAN concepts D7: Spaced revision, recovery and Month 3 preview

> **Gate:** score the Month 2 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 02 · Week 1 · daily cards

D029 · M02W1D1 · OSI seven-layer model and TCP/IP four-layer model

Phase: Academic month · **Optional date:** 2026-11-24 · **Domain resources:** N01 / N04

1. Day / sequence	Global Day 29 of 365; calendar-free M02W1D1; optional date 2026-11-24.
2. Exact topic	OSI seven-layer model and TCP/IP four-layer model
3. Exact subtopics	Physical, Data Link, Network, Transport, Session, Presentation, Application; Network Access, Internet, Transport, Application; encapsulation and decapsulation
4. Learning objective	Map a web request to each layer and name its protocol data unit
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 3, The TCP/IP Model; use the exact Day 3 lecture title on the playlist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a full encapsulation diagram from HTTP to bits and label PDUs at each layer; answer five mapping questions. **2.** Standard (~2 h): Draw a full encapsulation diagram from HTTP to bits and label PDUs at each layer; answer five mapping questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a full encapsulation diagram from HTTP to bits and label PDUs at each layer; answer five mapping questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	encapsulation.svg and answers.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Where is a MAC address used and where is an IP address used as data travels? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every layer has correct PDU, protocol and address type Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 3, The TCP/IP Model; use the exact Day 3 lecture title on the playlist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D030 · M02W1D2 · Ethernet frames, MAC addresses and switching

Phase: Academic month · **Optional date:** 2026-11-25 · **Domain resources:** N01 / N02

1. Day / sequence	Global Day 30 of 365; calendar-free M02W1D2; optional date 2026-11-25.
2. Exact topic	Ethernet frames, MAC addresses and switching
3. Exact subtopics	Ethernet header, FCS, MAC address format, unicast/broadcast, switch MAC learning, collision domains
4. Learning objective	Explain how a switch forwards a frame and why MAC addresses are local to a link
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 5 and Day 6, Ethernet LAN Switching Parts 1 and 2. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Index and Files `https://courses.jeremysitlab.com/p/ccna`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a small Packet Tracer LAN, inspect the switch MAC table before and after ping, and capture one frame if possible. **2.** Standard (~2 h): Build a small Packet Tracer LAN, inspect the switch MAC table before and after ping, and capture one frame if possible. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a small Packet Tracer LAN, inspect the switch MAC table before and after ping, and capture one frame if possible. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	switching-lab.pkt and annotated MAC table; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does the destination MAC change at each routed hop while IP endpoints may remain? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Switch table reflects learned addresses and the explanation is correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 5 and Day 6, Ethernet LAN Switching Parts 1 and 2"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D031 · M02W1D3 · ARP, ICMP and local address resolution

Phase: Academic month · **Optional date:** 2026-11-26 · **Domain resources:** N01 / N06

1. Day / sequence	Global Day 31 of 365; calendar-free M02W1D3; optional date 2026-11-26.
2. Exact topic	ARP, ICMP and local address resolution
3. Exact subtopics	ARP request/reply, ARP cache, gratuitous ARP, ICMP echo, TTL and basic error messages
4. Learning objective	Trace how a host finds the next-hop MAC address for an IP packet
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 12, Life of a Packet; Day 3 for ARP context. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate a ping on your own lab network or a Packet Tracer simulation; inspect ARP cache and packet sequence; explain each step. **2.** Standard (~2 h): Generate a ping on your own lab network or a Packet Tracer simulation; inspect ARP cache and packet sequence; explain each step. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate a ping on your own lab network or a Packet Tracer simulation; inspect ARP cache and packet sequence; explain each step. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	arp-icmp.md with Wireshark/Packet Tracer screenshots; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is ARP not routable and what security assumption does it make? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sequence is complete from ARP to ICMP reply and all addresses are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 12, Life of a Packet; Day 3 for ARP context"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D032 · M02W1D4 · VLANs, trunks and broadcast domains

Phase: Academic month · **Optional date:** 2026-11-27 · **Domain resources:** N01 / N03

1. Day / sequence	Global Day 32 of 365; calendar-free M02W1D4; optional date 2026-11-27.
2. Exact topic	VLANs, trunks and broadcast domains
3. Exact subtopics	Access ports, trunk ports, 802.1Q tags, native VLAN, broadcast domains and basic segmentation
4. Learning objective	Explain how VLANs isolate traffic and where tags appear
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 16-18, VLANs Parts 1-3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Files `https://jitl.jp/ccna-files`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure two VLANs in Packet Tracer, test intra/inter-VLAN behaviour, and document why a router or Layer 3 switch is needed between them. **2.** Standard (~2 h): Configure two VLANs in Packet Tracer, test intra/inter-VLAN behaviour, and document why a router or Layer 3 switch is needed between them. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure two VLANs in Packet Tracer, test intra/inter-VLAN behaviour, and document why a router or Layer 3 switch is needed between them. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vlan-lab.pkt and segmentation-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the security benefit and the main limitation of VLAN segmentation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Configuration works and the answer distinguishes isolation from strong security boundaries Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 16-18, VLANs Parts 1-3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D033 · M02W1D5 · IPv4 header, addresses and private ranges

Phase: Academic month · **Optional date:** 2026-11-28 · **Domain resources:** N01 / N11

1. Day / sequence	Global Day 33 of 365; calendar-free M02W1D5; optional date 2026-11-28.
2. Exact topic	IPv4 header, addresses and private ranges
3. Exact subtopics	IPv4 header fields, network/host portions, classful history, RFC1918 ranges, loopback and link-local addresses
4. Learning objective	Read an IPv4 header and identify private, public, loopback and link-local addresses
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 7-8 and Day 10, IPv4 Addressing Parts 1-2 and The IPv4 Header. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 791: Internet Protocol v4 `https://www.rfc-editor.org/rfc/rfc791`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Inspect an IPv4 packet in Wireshark; annotate version, IHL, DSCP, length, ID, flags, TTL, protocol, checksum and addresses. **2.** Standard (~2 h): Inspect an IPv4 packet in Wireshark; annotate version, IHL, DSCP, length, ID, flags, TTL, protocol, checksum and addresses. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Inspect an IPv4 packet in Wireshark; annotate version, IHL, DSCP, length, ID, flags, TTL, protocol, checksum and addresses. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ipv4-header.png with annotations; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is 192.168.1.10 not visible as a destination address on the public Internet? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All header fields are correctly mapped and address categories are identified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 7-8 and Day 10, IPv4 Addressing Parts 1-2 and The IPv4 Header"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D034 · M02W1D6 · CIDR, subnet masks and network/broadcast/host calculations

Phase: Academic month · **Optional date:** 2026-11-29 · **Domain resources:** N01 / N16

1. Day / sequence	Global Day 34 of 365; calendar-free M02W1D6; optional date 2026-11-29.
2. Exact topic	CIDR, subnet masks and network/broadcast/host calculations
3. Exact subtopics	Prefix length, wildcard mask, network address, broadcast address, first/last usable host, block size
4. Learning objective	Solve /8-/30 subnetting questions with a written method
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 13-15, Subnetting Parts 1-3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 20 timed subnetting questions; show binary boundaries for five; record mistakes and repeat each missed type. **2.** Standard (~2 h): Complete 20 timed subnetting questions; show binary boundaries for five; record mistakes and repeat each missed type. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 20 timed subnetting questions; show binary boundaries for five; record mistakes and repeat each missed type. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	subnetting-M2D6.csv with score and corrections; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Given 192.168.10.77/26, what are the network, broadcast and usable range? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least 80 percent accuracy with all corrections shown Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 13-15, Subnetting Parts 1-3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D035 · M02W1D7 · VLSM, route aggregation and daily subnetting fluency

Phase: Academic month · **Optional date:** 2026-11-30 · **Domain resources:** N17 / N16 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 35 of 365; calendar-free M02W1D7; optional date 2026-11-30.
2. Exact topic	VLSM, route aggregation and daily subnetting fluency
3. Exact subtopics	Variable-length subnet masks, route summarization, address conservation, longest-prefix match
4. Learning objective	Design an efficient addressing plan for a small topology
5. Primary lecture / reading + direct URL	Practical Networking Subnetting Mastery `https://www.practicalnetworking.net/stand-alone/subnetting-mastery/` **Type:** Course / articles · **Access:** Free articles/videos; optional paid resources
6. Exact section / lesson locator	Practical Networking Subnetting Mastery, parts on FLSM, VLSM and summarization. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Allocate subnets for five departments with different host counts, verify no overlap, then complete 20 timed questions — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Allocate subnets for five departments with different host counts, verify no overlap, then complete 20 timed questions — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Allocate subnets for five departments with different host counts, verify no overlap, then complete 20 timed questions — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vlsm-plan.csv and quiz score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which aggregate route covers your assigned subnets and what extra space does it include? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan has no overlap, meets host counts, and quiz accuracy is at least 80 percent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Practical Networking Subnetting Mastery, parts on FLSM, VLSM and summarization"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Subnetting Practice; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 · Week 2 · daily cards

D036 · M02W2D1 · Weekly packet and subnetting review

Phase: Academic month · **Optional date:** 2026-12-01 · **Domain resources:** N04 / N16

1. Day / sequence	Global Day 36 of 365; calendar-free M02W2D1; optional date 2026-12-01.
2. Exact topic	Weekly packet and subnetting review
3. Exact subtopics	OSI/TCP/IP, Ethernet, ARP, VLANs, IPv4, CIDR and VLSM; retrieval and error analysis
4. Learning objective	Consolidate Month 2 Week 1 and repair subnetting gaps
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009, objectives 1.1-1.4; use exact video titles on the course index. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 15 mixed questions, rework all mistakes, and write a one-page explanation of encapsulation with a subnetting example. **2.** Standard (~2 h): Complete 15 mixed questions, rework all mistakes, and write a one-page explanation of encapsulation with a subnetting example. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 15 mixed questions, rework all mistakes, and write a one-page explanation of encapsulation with a subnetting example. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2W1-review.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which layer or address changes at each hop and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70 percent and all corrections show working, not just answers Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009, objectives 1.1-1.4; use exact video titles on the course index"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D037 · M02W2D2 · Routing, default gateways and static routes

Phase: Academic month · **Optional date:** 2026-12-02 · **Domain resources:** N01 / N03

1. Day / sequence	Global Day 37 of 365; calendar-free M02W2D2; optional date 2026-12-02.
2. Exact topic	Routing, default gateways and static routes
3. Exact subtopics	Routing table, longest-prefix match, connected routes, default route, next hop, static route configuration
4. Learning objective	Predict and configure how a packet leaves its local network
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 11, Routing Fundamentals and Static Routing. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Files `https://jitl.jp/ccna-files`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a three-router Packet Tracer topology, configure static routes, and verify with ping/traceroute from both ends. **2.** Standard (~2 h): Build a three-router Packet Tracer topology, configure static routes, and verify with ping/traceroute from both ends. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a three-router Packet Tracer topology, configure static routes, and verify with ping/traceroute from both ends. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	routing-lab.pkt and route tables; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What happens when two routes match a destination but have different prefix lengths? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	End-to-end connectivity works and route selection is explained by longest-prefix match Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 11, Routing Fundamentals and Static Routing"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D038 · M02W2D3 · NAT, PAT and public versus private addressing

Phase: Academic month · **Optional date:** 2026-12-03 · **Domain resources:** N04 / N15

1. Day / sequence	Global Day 38 of 365; calendar-free M02W2D3; optional date 2026-12-03.
2. Exact topic	NAT, PAT and public versus private addressing
3. Exact subtopics	NAT translation table, inside/outside/local/global terminology, port address translation, limitations of NAT as security
4. Learning objective	Explain how many private hosts share one public address and why NAT is not a firewall
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 objective on NAT and port forwarding; use the course index to locate exact video. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Cisco Packet Tracer course page `https://www.netacad.com/courses/packet-tracer`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure NAT/PAT in Packet Tracer or diagram a translation table for five local connections; test one external connection. **2.** Standard (~2 h): Configure NAT/PAT in Packet Tracer or diagram a translation table for five local connections; test one external connection. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure NAT/PAT in Packet Tracer or diagram a translation table for five local connections; test one external connection. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	nat-table.csv and topology diagram; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does NAT not guarantee inbound protection or application-layer security? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Translation entries correctly map addresses/ports and limitations are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 objective on NAT and port forwarding; use the course index to locate exact video"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D039 · M02W2D4 · Dynamic routing and routing security concepts

Phase: Academic month · **Optional date:** 2026-12-04 · **Domain resources:** N01 / N02

1. Day / sequence	Global Day 39 of 365; calendar-free M02W2D4; optional date 2026-12-04.
2. Exact topic	Dynamic routing and routing security concepts
3. Exact subtopics	OSPF areas and adjacencies, metrics, route propagation, authentication, route filtering and failure domains
4. Learning objective	Explain the difference between static and dynamic routing and one security risk of incorrect route advertisement
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 24-28, Dynamic Routing through OSPF Part 3; select the Day 26 lecture for a first pass. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Index and Files `https://courses.jeremysitlab.com/p/ccna`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure single-area OSPF in Packet Tracer, break one link, observe convergence, and write a security caveat. **2.** Standard (~2 h): Configure single-area OSPF in Packet Tracer, break one link, observe convergence, and write a security caveat. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure single-area OSPF in Packet Tracer, break one link, observe convergence, and write a security caveat. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ospf-lab.pkt and convergence notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can an unauthorized routing update redirect traffic even without exploiting an application? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Neighborship forms, connectivity restores, and the risk is correctly described Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 24-28, Dynamic Routing through OSPF Part 3; select the Day 26 lecture for a first pass"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D040 · M02W2D5 · TCP handshake, sequence numbers and reliability

Phase: Academic month · **Optional date:** 2026-12-05 · **Domain resources:** N01 / N10

1. Day / sequence	Global Day 40 of 365; calendar-free M02W2D5; optional date 2026-12-05.
2. Exact topic	TCP handshake, sequence numbers and reliability
3. Exact subtopics	SYN, SYN-ACK, ACK; sequence and acknowledgment numbers; windowing, retransmission, FIN/RST, TCP state
4. Learning objective	Annotate a TCP connection and explain how reliability emerges from sequence numbers
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 30, TCP and UDP. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 9293: Transmission Control Protocol `https://www.rfc-editor.org/rfc/rfc9293`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Capture a local TCP connection with Wireshark, label handshake, seq/ack progression, one data segment and teardown. **2.** Standard (~2 h): Capture a local TCP connection with Wireshark, label handshake, seq/ack progression, one data segment and teardown. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Capture a local TCP connection with Wireshark, label handshake, seq/ack progression, one data segment and teardown. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	tcp-handshake.png and field table; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is the initial sequence number unpredictable and security-relevant? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All packet fields are correctly labelled and the reliability explanation is coherent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 30, TCP and UDP"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D041 · M02W2D6 · UDP, ports and common services

Phase: Academic month · **Optional date:** 2026-12-06 · **Domain resources:** N01 / N14

1. Day / sequence	Global Day 41 of 365; calendar-free M02W2D6; optional date 2026-12-06.
2. Exact topic	UDP, ports and common services
3. Exact subtopics	UDP header, connectionless behaviour, DNS/DHCP/QUIC use cases, ephemeral ports, TCP versus UDP trade-offs
4. Learning objective	Decide when an application should use UDP and explain the security implications
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 30, TCP and UDP. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	IANA Service Name and Transport Protocol Port Number Registry `https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Capture a DNS query over UDP, compare its header to TCP, and build a table of ten common ports with protocol and purpose. **2.** Standard (~2 h): Capture a DNS query over UDP, compare its header to TCP, and build a table of ten common ports with protocol and purpose. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Capture a DNS query over UDP, compare its header to TCP, and build a table of ten common ports with protocol and purpose. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	udp-dns.md and ports.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is UDP spoofing often easier than TCP spoofing? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Capture and port table are correct and the trade-off explanation includes integrity/authentication Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 30, TCP and UDP"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D042 · M02W2D7 · DNS resolution and record types

Phase: Academic month · **Optional date:** 2026-12-07 · **Domain resources:** N13 / N01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 42 of 365; calendar-free M02W2D7; optional date 2026-12-07.
2. Exact topic	DNS resolution and record types
3. Exact subtopics	Recursive/iterative lookup, root/TLD/authoritative servers, A/AAAA/CNAME/MX/TXT/NS/SOA, TTL, caching and DNSSEC concepts
4. Learning objective	Trace a DNS answer and explain why DNS is both naming and a security control point
5. Primary lecture / reading + direct URL	Cloudflare Learning Center: What is DNS? `https://www.cloudflare.com/learning/dns/what-is-dns/` **Type:** Educational documentation · **Access:** Free
6. Exact section / lesson locator	Cloudflare Learning Center, What is DNS and DNS server/resolution sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Use dig or nslookup to query an example domain from your own machine, capture the exchange, and record each answer section and TTL — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Use dig or nslookup to query an example domain from your own machine, capture the exchange, and record each answer section and TTL — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Use dig or nslookup to query an example domain from your own machine, capture the exchange, and record each answer section and TTL — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	dns-trace.txt and annotated capture; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which party is authoritative for a name and why can a cached answer be stale? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Resolution chain is correct and capture fields are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Cloudflare Learning Center, What is DNS and DNS server/resolution sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Suricata documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 · Week 3 · daily cards

D043 · M02W3D1 · DHCP, DHCPv6 and network configuration lifecycle

Phase: Academic month · **Optional date:** 2026-12-08 · **Domain resources:** N04 / N05

1. Day / sequence	Global Day 43 of 365; calendar-free M02W3D1; optional date 2026-12-08.
2. Exact topic	DHCP, DHCPv6 and network configuration lifecycle
3. Exact subtopics	DORA process, lease, scope, options, default gateway, DNS server, reservation, rogue DHCP risk
4. Learning objective	Explain how a host obtains configuration and what an attacker could do with a rogue DHCP server
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 objective on DHCP; use the exact objective video title in the course index. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer N10-009 Course Index `https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Observe your own lab VM's DHCP lease and options; diagram the four-message exchange; list two mitigations for rogue DHCP. **2.** Standard (~2 h): Observe your own lab VM's DHCP lease and options; diagram the four-message exchange; list two mitigations for rogue DHCP. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Observe your own lab VM's DHCP lease and options; diagram the four-message exchange; list two mitigations for rogue DHCP. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	dhcp-lease.txt and DORA diagram; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why are DHCP options security-relevant beyond assigning an IP address? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Observed lease fields are identified and mitigations are technically plausible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 objective on DHCP; use the exact objective video title in the course index"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D044 · M02W3D2 · HTTP, HTTPS and TLS at the network layer

Phase: Academic month · **Optional date:** 2026-12-09 · **Domain resources:** F04 / C13

1. Day / sequence	Global Day 44 of 365; calendar-free M02W3D2; optional date 2026-12-09.
2. Exact topic	HTTP, HTTPS and TLS at the network layer
3. Exact subtopics	HTTP over TCP, TLS handshake, certificates, ALPN, HTTPS port 443, encrypted versus visible metadata, HSTS
4. Learning objective	Explain what TLS protects and what metadata remains visible
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 2 Securing Systems, HTTPS/TLS/certificates notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 8446: TLS 1.3 `https://www.rfc-editor.org/rfc/rfc8446`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Capture a browser connection to an authorized site or a local HTTPS test server; identify ClientHello, certificate, encrypted application data and visible IP/DNS metadata. **2.** Standard (~2 h): Capture a browser connection to an authorized site or a local HTTPS test server; identify ClientHello, certificate, encrypted application data and visible IP/DNS metadata. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Capture a browser connection to an authorized site or a local HTTPS test server; identify ClientHello, certificate, encrypted application data and visible IP/DNS metadata. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	tls-capture.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which parts of HTTPS are visible to a network observer and which are protected? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Capture is correctly annotated and the answer distinguishes encryption from anonymity Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 2 Securing Systems, HTTPS/TLS/certificates notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D045 · M02W3D3 · Weekly network services practical and retrieval

Phase: Academic month · **Optional date:** 2026-12-10 · **Domain resources:** N01 / N16

1. Day / sequence	Global Day 45 of 365; calendar-free M02W3D3; optional date 2026-12-10.
2. Exact topic	Weekly network services practical and retrieval
3. Exact subtopics	Routing, NAT, TCP, UDP, DNS, DHCP and TLS; 20 subnetting questions; service troubleshooting
4. Learning objective	Diagnose a broken small network using the OSI layer method
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 38, DNS, and Day 30, TCP and UDP, for recap. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In Packet Tracer or a VM lab, fix three injected faults and complete 20 timed subnetting questions; write a layer-by-layer diagnosis. **2.** Standard (~2 h): In Packet Tracer or a VM lab, fix three injected faults and complete 20 timed subnetting questions; write a layer-by-layer diagnosis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In Packet Tracer or a VM lab, fix three injected faults and complete 20 timed subnetting questions; write a layer-by-layer diagnosis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2W2-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence allowed you to eliminate layers 1, 2 and 3 before checking DNS? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All faults are fixed, diagnosis is evidence-based, and subnetting score is at least 80 percent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 38, DNS, and Day 30, TCP and UDP, for recap"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D046 · M02W3D4 · Firewalls, proxies, VPNs and segmentation architectures

Phase: Academic month · **Optional date:** 2026-12-11 · **Domain resources:** C01 / C06

1. Day / sequence	Global Day 46 of 365; calendar-free M02W3D4; optional date 2026-12-11.
2. Exact topic	Firewalls, proxies, VPNs and segmentation architectures
3. Exact subtopics	Stateless/stateful rules, default deny, DMZ, forward/reverse proxies, VPN tunnels, microsegmentation and zero-trust concepts
4. Learning objective	Compare controls by what they inspect, where they operate and what they cannot see
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 Security+ objective 3.2 and 4.4 videos on network security and segmentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw three architectures for a small fictional organization and write a rule table for each; identify one bypass condition. **2.** Standard (~2 h): Draw three architectures for a small fictional organization and write a rule table for each; identify one bypass condition. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw three architectures for a small fictional organization and write a rule table for each; identify one bypass condition. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	firewall-architectures.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a VPN not automatically make a compromised laptop trustworthy? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each control's scope and limitation is explicit and rule tables are internally consistent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 Security+ objective 3.2 and 4.4 videos on network security and segmentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D047 · M02W3D5 · Wireshark display filters and protocol dissection

Phase: Academic month · **Optional date:** 2026-12-12 · **Domain resources:** N06 / N07

1. Day / sequence	Global Day 47 of 365; calendar-free M02W3D5; optional date 2026-12-12.
2. Exact topic	Wireshark display filters and protocol dissection
3. Exact subtopics	Capture filters versus display filters, conversations, endpoints, Follow TCP Stream, protocol hierarchy and expert info
4. Learning objective	Use Wireshark to answer questions from a legal sample capture without guessing
5. Primary lecture / reading + direct URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Wireshark User's Guide, Chapters 4-6: capturing, display filters and packet dissection. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Open a Wireshark sample HTTP/DNS capture; list top talkers, identify request/response pairs and export one structured field table. **2.** Standard (~2 h): Open a Wireshark sample HTTP/DNS capture; list top talkers, identify request/response pairs and export one structured field table. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Open a Wireshark sample HTTP/DNS capture; list top talkers, identify request/response pairs and export one structured field table. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	wireshark-M2D17.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which display filter isolates DNS responses and how is it different from a capture filter? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All questions are answered with packet-level evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark User's Guide, Chapters 4-6: capturing, display filters and packet dissection"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark SampleCaptures; write a five-line correction; retest next morning with a new 10-minute problem.

D048 · M02W3D6 · Nmap host discovery and port scanning in an isolated lab

Phase: Academic month · **Optional date:** 2026-12-13 · **Domain resources:** N09 / N08

1. Day / sequence	Global Day 48 of 365; calendar-free M02W3D6; optional date 2026-12-13.
2. Exact topic	Nmap host discovery and port scanning in an isolated lab
3. Exact subtopics	Written scope, target specification, host discovery, TCP SYN scan, connect scan, version detection, timing and output formats
4. Learning objective	Interpret scan results and explain how each probe appears on the wire
5. Primary lecture / reading + direct URL	Nmap Reference Guide `https://nmap.org/book/man.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Nmap Reference Guide, host discovery, port scanning techniques and service/version detection sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Scan only a dedicated lab VM you own; save normal, XML and grepable output; capture packets during a small scan and map probes to results. **2.** Standard (~2 h): Scan only a dedicated lab VM you own; save normal, XML and grepable output; capture packets during a small scan and map probes to results. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Scan only a dedicated lab VM you own; save normal, XML and grepable output; capture packets during a small scan and map probes to results. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	scope.txt, nmap outputs and packet capture; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a closed port reply RST while a filtered port produce no response? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Written scope exists, targets are private/isolated, and results are reconciled with the capture Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nmap Reference Guide, host discovery, port scanning techniques and service/version detection sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RFC 9293: Transmission Control Protocol; write a five-line correction; retest next morning with a new 10-minute problem.

D049 · M02W3D7 · TCP retransmissions, MTU and fragmentation analysis

Phase: Academic month · **Optional date:** 2026-12-14 · **Domain resources:** N10 / N07 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 49 of 365; calendar-free M02W3D7; optional date 2026-12-14.
2. Exact topic	TCP retransmissions, MTU and fragmentation analysis
3. Exact subtopics	Retransmission, duplicate ACK, window size, MTU, MSS, fragmentation flags and Path MTU Discovery
4. Learning objective	Diagnose slow or broken connections from packet-level symptoms
5. Primary lecture / reading + direct URL	RFC 9293: Transmission Control Protocol `https://www.rfc-editor.org/rfc/rfc9293` **Type:** Authoritative standard · **Access:** Free
6. Exact section / lesson locator	RFC 9293 sections on retransmission and flow control; use Wireshark's expert info alongside the standard. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Analyze a provided sample capture with retransmissions; calculate one round-trip estimate and identify fragmentation or MTU clues — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Analyze a provided sample capture with retransmissions; calculate one round-trip estimate and identify fragmentation or MTU clues — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Analyze a provided sample capture with retransmissions; calculate one round-trip estimate and identify fragmentation or MTU clues — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	retransmission-analysis.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: How would you distinguish congestion loss from an MTU black hole using packet evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Symptoms are supported by specific frames and confidence limits are stated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "RFC 9293 sections on retransmission and flow control; use Wireshark's expert info alongside the standard"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Suricata documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 · Week 4 · daily cards

D050 · M02W4D1 · Network troubleshooting methodology and tool selection

Phase: Academic month · **Optional date:** 2026-12-15 · **Domain resources:** N05 / N08

1. Day / sequence	Global Day 50 of 365; calendar-free M02W4D1; optional date 2026-12-15.
2. Exact topic	Network troubleshooting methodology and tool selection
3. Exact subtopics	Layered hypothesis testing, ipconfig/ip/ifconfig, ping, traceroute, ss/netstat, dig, curl, logs, baseline and rollback
4. Learning objective	Resolve an unknown network issue with minimal changes and documented evidence
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Course Index `https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/` **Type:** Course index · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 course index, Network Troubleshooting objective videos. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Diagnose three instructor-created or self-created faults in the isolated lab; record command, prediction, result and conclusion for each step. **2.** Standard (~2 h): Diagnose three instructor-created or self-created faults in the isolated lab; record command, prediction, result and conclusion for each step. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Diagnose three instructor-created or self-created faults in the isolated lab; record command, prediction, result and conclusion for each step. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	troubleshooting-M2D20.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which observation changed your hypothesis and what alternative did it eliminate? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All faults are resolved and each step includes a prediction and result Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 course index, Network Troubleshooting objective videos"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Network+ Training Course; write a five-line correction; retest next morning with a new 10-minute problem.

D051 · M02W4D2 · Network-security project: topology and traffic report

Phase: Academic month · **Optional date:** 2026-12-16 · **Domain resources:** N06 / N09

1. Day / sequence	Global Day 51 of 365; calendar-free M02W4D2; optional date 2026-12-16.
2. Exact topic	Network-security project: topology and traffic report
3. Exact subtopics	Asset inventory, trust boundaries, traffic flows, segmentation, capture evidence, risks and mitigations
4. Learning objective	Produce a professional network diagram and traffic-analysis report
5. Primary lecture / reading + direct URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Wireshark User's Guide statistics and protocol hierarchy chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Nmap Reference Guide `https://nmap.org/book/man.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Combine your lab topology, one packet capture, one authorized scan and a segmentation proposal into a single report. **2.** Standard (~2 h): Combine your lab topology, one packet capture, one authorized scan and a segmentation proposal into a single report. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Combine your lab topology, one packet capture, one authorized scan and a segmentation proposal into a single report. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-03 and PRJ-04 evidence package; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which traffic flow is riskiest and what control would reduce it without breaking function? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram, capture, scan and report are consistent and all identifiers are sanitized Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark User's Guide statistics and protocol hierarchy chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark SampleCaptures; write a five-line correction; retest next morning with a new 10-minute problem.

D052 · M02W4D3 · Monthly network practical and daily subnetting test

Phase: Academic month · **Optional date:** 2026-12-17 · **Domain resources:** N01 / N16

1. Day / sequence	Global Day 52 of 365; calendar-free M02W4D3; optional date 2026-12-17.
2. Exact topic	Monthly network practical and daily subnetting test
3. Exact subtopics	OSI/TCPIP, Ethernet, ARP, VLANs, IPv4/CIDR, routing, NAT, TCP/UDP, DNS/DHCP/TLS, Wireshark, Nmap and troubleshooting
4. Learning objective	Demonstrate independent networking competence under timed conditions
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Course Index Days 1-30 for targeted revision; use exact Day titles only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute practical with 20 subnetting questions, one unfamiliar capture, one broken topology and one scope assessment. **2.** Standard (~2 h): Complete a 90-minute practical with 20 subnetting questions, one unfamiliar capture, one broken topology and one scope assessment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute practical with 20 subnetting questions, one unfamiliar capture, one broken topology and one scope assessment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2-practical.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you prove a DNS issue rather than a routing issue to a skeptical teammate? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Course Index Days 1-30 for targeted revision; use exact Day titles only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D053 · M02W4D4 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2026-12-18 · **Domain resources:** N04 / N17

1. Day / sequence	Global Day 53 of 365; calendar-free M02W4D4; optional date 2026-12-18.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Error-log analysis, subnetting weak types, packet-analysis mistakes, firewall misconceptions and project fixes
4. Learning objective	Close Month 2 gaps and produce a prerequisites checklist for Month 3
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 Course Index, objectives 1-5; choose only the three weakest objectives. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Practical Networking Subnetting Mastery `https://www.practicalnetworking.net/stand-alone/subnetting-mastery/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework every failed assessment item with a new example; revisit the matching video; retake 20 subnetting questions; update project README. **2.** Standard (~2 h): Rework every failed assessment item with a new example; revisit the matching video; retake 20 subnetting questions; update project README. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework every failed assessment item with a new example; revisit the matching video; retake 20 subnetting questions; update project README. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2-review.md and retest score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which three network concepts are prerequisites for Linux and Windows security? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and each original gap has new evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 Course Index, objectives 1-5; choose only the three weakest objectives"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D054 · M02W4D5 · IPv6 addressing and protocol differences

Phase: Academic month · **Optional date:** 2026-12-19 · **Domain resources:** N01 / N12

1. Day / sequence	Global Day 54 of 365; calendar-free M02W4D5; optional date 2026-12-19.
2. Exact topic	IPv6 addressing and protocol differences
3. Exact subtopics	IPv6 address notation, prefix, global/unicast/link-local, EUI-64/privacy addresses, ICMPv6, neighbour discovery and extension headers
4. Learning objective	Compare IPv4 and IPv6 address configuration and security assumptions
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 31-33, IPv6 Parts 1-3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 8200: Internet Protocol v6 `https://www.rfc-editor.org/rfc/rfc8200`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write ten IPv6 addresses in compressed and expanded form, configure a lab IPv6 address if supported, and compare an IPv6 header with IPv4. **2.** Standard (~2 h): Write ten IPv6 addresses in compressed and expanded form, configure a lab IPv6 address if supported, and compare an IPv6 header with IPv4. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write ten IPv6 addresses in compressed and expanded form, configure a lab IPv6 address if supported, and compare an IPv6 header with IPv4. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ipv6-notes.md and optional Packet Tracer file; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does IPv6 still need firewall policy despite its larger address space? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Address conversions are correct and the security answer is not based on address obscurity Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 31-33, IPv6 Parts 1-3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D055 · M02W4D6 · Wireless LANs, network monitoring and SPAN concepts

Phase: Academic month · **Optional date:** 2026-12-20 · **Domain resources:** N01 / N06

1. Day / sequence	Global Day 55 of 365; calendar-free M02W4D6; optional date 2026-12-20.
2. Exact topic	Wireless LANs, network monitoring and SPAN concepts
3. Exact subtopics	802.11 association, SSID/BSSID, WPA2/WPA3 concepts, channel, switch port mirroring/SPAN, taps and flow records
4. Learning objective	Explain where to place a sensor and what it can and cannot observe
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA wireless lectures in the Day 50+ sequence; use the course index to locate current Day numbers. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw two monitoring placements for a small network, list visible/invisible traffic for each, and state privacy/authorization constraints. **2.** Standard (~2 h): Draw two monitoring placements for a small network, list visible/invisible traffic for each, and state privacy/authorization constraints. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw two monitoring placements for a small network, list visible/invisible traffic for each, and state privacy/authorization constraints. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	monitoring-design.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does encrypted traffic still yield useful metadata for defenders? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Design correctly addresses span, taps, encryption visibility and legal authorization Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA wireless lectures in the Day 50+ sequence; use the course index to locate current Day numbers"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D056 · M02W4D7 · Spaced revision, recovery and Month 3 preview

Phase: Academic month · **Optional date:** 2026-12-21 · **Domain resources:** N16 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 56 of 365; calendar-free M02W4D7; optional date 2026-12-21.
2. Exact topic	Spaced revision, recovery and Month 3 preview
3. Exact subtopics	Retrieval practice, command notebook cleanup, OSI mapping, binary/CIDR fluency and lab snapshots
4. Learning objective	Preserve networking knowledge and prepare Linux/Windows system administration
5. Primary lecture / reading + direct URL	Subnetting Practice `https://subnettingpractice.com/` **Type:** Interactive practice · **Access:** Free tier; optional ads/premium features may appear
6. Exact section / lesson locator	Subnetting Practice mixed set; 15-question maintenance drill. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create ten flashcards from errors, clean VM snapshots, install/verify Linux and Windows lab tools, and rest or do a 30-minute light review — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create ten flashcards from errors, clean VM snapshots, install/verify Linux and Windows lab tools, and rest or do a 30-minute light review — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create ten flashcards from errors, clean VM snapshots, install/verify Linux and Windows lab tools, and rest or do a 30-minute light review — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Explain one full packet journey from DNS request to TLS session in ten sentences without notes Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Snapshot is clean, flashcards are scheduled, and the explanation is complete Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Subnetting Practice mixed set; 15-question maintenance drill"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Practical Networking Subnetting Mastery; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 — Linux, Windows, system administration and hardening

Days D057-D084. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Linux filesystem hierarchy and package management D2: Linux users, groups, sudo and file permissions D3: Processes, services and resource monitoring D4: Bash scripting fundamentals and quoting D5: Pipes, redirection and text processing D6: SSH keys, remote administration and secure configuration D7: Weekly Linux review and Bandit independent practice
Week 2	D1: Linux networking and firewall basics D2: Linux logs, auditd concepts and evidence integrity D3: Linux hardening and CIS controls D4: Systemd security and privilege boundaries D5: OverTheWire Bandit intermediate independent lab D6: Weekly Linux practical and troubleshooting D7: Windows architecture, accounts and permissions
Week 3	D1: Windows processes, services and Sysinternals D2: PowerShell fundamentals and safe scripting D3: Windows Event Viewer and event IDs D4: Sysmon installation and process telemetry D5: Active Directory architecture and identity boundaries D6: Weekly Windows and AD review D7: Group Policy and security baselines
Week 4	D1: Endpoint security controls and patch management D2: Windows image and event-log investigation project D3: Linux and Windows audit utility project D4: Month 3 practical examination and milestone gate D5: Monthly review, cleanup and Month 4 preview D6: Kerberos, NTLM and AD authentication flows D7: Spaced revision, recovery and portfolio update

> **Gate:** score the Month 3 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 03 · Week 1 · daily cards

D057 · M03W1D1 · Linux filesystem hierarchy and package management

Phase: Academic month · **Optional date:** 2026-12-22 · **Domain resources:** L01 / F13

1. Day / sequence	Global Day 57 of 365; calendar-free M03W1D1; optional date 2026-12-22.
2. Exact topic	Linux filesystem hierarchy and package management
3. Exact subtopics	FHS, absolute/relative paths, package repositories, apt, dpkg, dependencies and update hygiene
4. Learning objective	Navigate Linux and manage packages from the CLI with a security rationale
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Filesystem Hierarchy and Package Management sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Ubuntu Desktop Guide `https://help.ubuntu.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In the Ubuntu VM, locate ten configuration paths, install one benign package, inspect its files, then remove it and verify cleanup. **2.** Standard (~2 h): In the Ubuntu VM, locate ten configuration paths, install one benign package, inspect its files, then remove it and verify cleanup. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In the Ubuntu VM, locate ten configuration paths, install one benign package, inspect its files, then remove it and verify cleanup. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-inventory.md and package transcript; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should you avoid installing random security tools from untrusted repositories? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All paths are correctly explained and package state is verified before/after Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Filesystem Hierarchy and Package Management sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D058 · M03W1D2 · Linux users, groups, sudo and file permissions

Phase: Academic month · **Optional date:** 2026-12-23 · **Domain resources:** L01 / L04

1. Day / sequence	Global Day 58 of 365; calendar-free M03W1D2; optional date 2026-12-23.
2. Exact topic	Linux users, groups, sudo and file permissions
3. Exact subtopics	/etc/passwd, /etc/shadow, /etc/group, UID/GID, umask, setuid/setgid, sticky bit, sudoers
4. Learning objective	Audit permissions and apply least privilege safely
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Users and Groups and Permissions sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Find setuid binaries in the VM, explain three of them, create a shared group directory, and check sudo configuration without editing it blindly. **2.** Standard (~2 h): Find setuid binaries in the VM, explain three of them, create a shared group directory, and check sudo configuration without editing it blindly. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Find setuid binaries in the VM, explain three of them, create a shared group directory, and check sudo configuration without editing it blindly. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-permissions.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a world-writable setuid binary dangerous? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Audit is complete and no insecure test permissions remain Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Users and Groups and Permissions sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D059 · M03W1D3 · Processes, services and resource monitoring

Phase: Academic month · **Optional date:** 2026-12-24 · **Domain resources:** L05 / L04

1. Day / sequence	Global Day 59 of 365; calendar-free M03W1D3; optional date 2026-12-24.
2. Exact topic	Processes, services and resource monitoring
3. Exact subtopics	ps, top, htop if available, nice, kill, systemctl, journalctl, cgroups and service dependencies
4. Learning objective	Explain process trees and control a benign service without harming the host
5. Primary lecture / reading + direct URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	systemd.unit manual, unit dependencies and targets. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Start, stop, restart, enable and disable a harmless service; inspect logs and CPU/memory; map parent/child processes. **2.** Standard (~2 h): Start, stop, restart, enable and disable a harmless service; inspect logs and CPU/memory; map parent/child processes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Start, stop, restart, enable and disable a harmless service; inspect logs and CPU/memory; map parent/child processes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-processes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between stopping and disabling a service? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Commands, unit state and logs agree with the written explanation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "systemd.unit manual, unit dependencies and targets"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux man-pages project; write a five-line correction; retest next morning with a new 10-minute problem.

D060 · M03W1D4 · Bash scripting fundamentals and quoting

Phase: Academic month · **Optional date:** 2026-12-25 · **Domain resources:** L03 / L01

1. Day / sequence	Global Day 60 of 365; calendar-free M03W1D4; optional date 2026-12-25.
2. Exact topic	Bash scripting fundamentals and quoting
3. Exact subtopics	Shebang, variables, expansion, quoting, exit codes, conditionals, loops, functions, getopts, shellcheck
4. Learning objective	Write a defensive script that fails safely on malformed input
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	Greg's Wiki BashGuide, Commands and Arguments, Quoting, Tests and Loops. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create an inventory script that collects hostname, kernel, users and listening ports; handle empty fields and filenames with spaces. **2.** Standard (~2 h): Create an inventory script that collects hostname, kernel, users and listening ports; handle empty fields and filenames with spaces. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create an inventory script that collects hostname, kernel, users and listening ports; handle empty fields and filenames with spaces. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	audit.sh and shellcheck output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does unquoted variable expansion break filenames with spaces and create security bugs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Script runs with set -euo pipefail where appropriate and passes shellcheck for assigned issues Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Greg's Wiki BashGuide, Commands and Arguments, Quoting, Tests and Loops"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D061 · M03W1D5 · Pipes, redirection and text processing

Phase: Academic month · **Optional date:** 2026-12-26 · **Domain resources:** L03 / P04

1. Day / sequence	Global Day 61 of 365; calendar-free M03W1D5; optional date 2026-12-26.
2. Exact topic	Pipes, redirection and text processing
3. Exact subtopics	stdin/stdout/stderr, pipes, grep, sed, awk, cut, sort, uniq, tee, regular expressions, process substitution
4. Learning objective	Compose safe command pipelines to analyze synthetic logs
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	Greg's Wiki BashGuide, Redirection and Arrays; P04 RegexOne Lessons 1-8. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RegexOne `https://regexone.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a pipeline that counts failed SSH logins by user/IP from a synthetic auth.log and explain each stage. **2.** Standard (~2 h): Build a pipeline that counts failed SSH logins by user/IP from a synthetic auth.log and explain each stage. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a pipeline that counts failed SSH logins by user/IP from a synthetic auth.log and explain each stage. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	log-pipeline.sh and results.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should script input be treated as untrusted even if it came from a local log? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline output is verified against a hand-calculated sample Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Greg's Wiki BashGuide, Redirection and Arrays; P04 RegexOne Lessons 1-8"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D062 · M03W1D6 · SSH keys, remote administration and secure configuration

Phase: Academic month · **Optional date:** 2026-12-27 · **Domain resources:** L01 / C08

1. Day / sequence	Global Day 62 of 365; calendar-free M03W1D6; optional date 2026-12-27.
2. Exact topic	SSH keys, remote administration and secure configuration
3. Exact subtopics	SSH protocol roles, host keys, key pairs, authorized_keys, sshd_config basics, known_hosts and MFA concept
4. Learning objective	Set up key-based SSH between two lab VMs without exposing services publicly
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, SSH and Networking sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-63B `https://pages.nist.gov/800-63-4/sp800-63b.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate an ed25519 key pair, install the public key, verify host-key fingerprint, disable password login only in a disposable VM, and document rollback. **2.** Standard (~2 h): Generate an ed25519 key pair, install the public key, verify host-key fingerprint, disable password login only in a disposable VM, and document rollback. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate an ed25519 key pair, install the public key, verify host-key fingerprint, disable password login only in a disposable VM, and document rollback. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ssh-lab.md and config diffs; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What does a host-key change warning mean and when could it be legitimate? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Key-based login works, host key is verified, and no production or host networking is exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, SSH and Networking sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D063 · M03W1D7 · Weekly Linux review and Bandit independent practice

Phase: Academic month · **Optional date:** 2026-12-28 · **Domain resources:** L02 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 63 of 365; calendar-free M03W1D7; optional date 2026-12-28.
2. Exact topic	Weekly Linux review and Bandit independent practice
3. Exact subtopics	Filesystem, permissions, processes, Bash, pipes, SSH and troubleshooting; active recall
4. Learning objective	Demonstrate independent Linux problem-solving without walkthroughs
5. Primary lecture / reading + direct URL	OverTheWire Bandit `https://overthewire.org/wargames/bandit/` **Type:** CTF / command-line lab · **Access:** Free
6. Exact section / lesson locator	OverTheWire Bandit Levels 0-10; use only the official level pages and man pages for hints. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve at least six Bandit levels independently, write concept notes rather than passwords, and complete a 15-question retrieval quiz — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve at least six Bandit levels independently, write concept notes rather than passwords, and complete a 15-question retrieval quiz — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve at least six Bandit levels independently, write concept notes rather than passwords, and complete a 15-question retrieval quiz — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	bandit-notes.md and weekly-review-M3W1.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which command revealed the crucial fact in each level and why did it work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Six levels are solved with original notes and quiz score is at least 70 percent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OverTheWire Bandit Levels 0-10; use only the official level pages and man pages for hints"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux Journey; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 · Week 2 · daily cards

D064 · M03W2D1 · Linux networking and firewall basics

Phase: Academic month · **Optional date:** 2026-12-29 · **Domain resources:** L01 / N08

1. Day / sequence	Global Day 64 of 365; calendar-free M03W2D1; optional date 2026-12-29.
2. Exact topic	Linux networking and firewall basics
3. Exact subtopics	ip addr/route/link, ss, resolvectl, ufw/nftables concepts, /etc/hosts, DNS resolution and loopback
4. Learning objective	Inspect and control local Linux network state safely
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Networking sections; L04 man pages for ip and ss. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure a static address on an internal adapter in the VM, inspect routes and listening ports, apply one harmless firewall rule and test connectivity. **2.** Standard (~2 h): Configure a static address on an internal adapter in the VM, inspect routes and listening ports, apply one harmless firewall rule and test connectivity. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure a static address on an internal adapter in the VM, inspect routes and listening ports, apply one harmless firewall rule and test connectivity. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-network.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a service be running but unreachable even when the firewall allows the port? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Network state, process state and connectivity evidence are all consistent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Networking sections; L04 man pages for ip and ss"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D065 · M03W2D2 · Linux logs, auditd concepts and evidence integrity

Phase: Academic month · **Optional date:** 2026-12-30 · **Domain resources:** L09 / L05

1. Day / sequence	Global Day 65 of 365; calendar-free M03W2D2; optional date 2026-12-30.
2. Exact topic	Linux logs, auditd concepts and evidence integrity
3. Exact subtopics	/var/log, journald, auth.log, syslog, log rotation, timestamps, time zones, hashes and chain of custody
4. Learning objective	Collect and preserve Linux logs without contaminating evidence
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation for comparison; L04 man pages for journalctl and logger. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate benign failed logins, locate them in two log sources, hash the exported evidence and write an evidence-handling note. **2.** Standard (~2 h): Generate benign failed logins, locate them in two log sources, hash the exported evidence and write an evidence-handling note. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate benign failed logins, locate them in two log sources, hash the exported evidence and write an evidence-handling note. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-log-evidence/ with SHA-256 manifest; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is time synchronization important for correlating logs from two hosts? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Exported logs are unchanged per hash and the note includes source, time and collector Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation for comparison; L04 man pages for journalctl and logger"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D066 · M03W2D3 · Linux hardening and CIS controls

Phase: Academic month · **Optional date:** 2026-12-31 · **Domain resources:** L14 / L13

1. Day / sequence	Global Day 66 of 365; calendar-free M03W2D3; optional date 2026-12-31.
2. Exact topic	Linux hardening and CIS controls
3. Exact subtopics	Patch management, service minimization, SSH hardening, password policy, audit, firewall, file integrity and backups
4. Learning objective	Apply and validate five low-risk hardening controls in a snapshot
5. Primary lecture / reading + direct URL	CIS Benchmarks `https://www.cisecurity.org/cis-benchmarks` **Type:** Consensus benchmarks · **Access:** Free PDFs may require registration; some tools paid
6. Exact section / lesson locator	CIS Benchmarks, Ubuntu or Debian profile; select five Level 1 controls. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Windows security baselines `https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Take a snapshot, record baseline, apply five controls, test functionality, then restore one control and document impact. **2.** Standard (~2 h): Take a snapshot, record baseline, apply five controls, test functionality, then restore one control and document impact. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Take a snapshot, record baseline, apply five controls, test functionality, then restore one control and document impact. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	hardening-baseline.md and before/after evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control created a compatibility problem and how would you communicate that risk? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each control has a test, result and rollback note; VM remains functional Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CIS Benchmarks, Ubuntu or Debian profile; select five Level 1 controls"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Windows security baselines; write a five-line correction; retest next morning with a new 10-minute problem.

D067 · M03W2D4 · Systemd security and privilege boundaries

Phase: Academic month · **Optional date:** 2027-01-01 · **Domain resources:** L05 / L04

1. Day / sequence	Global Day 67 of 365; calendar-free M03W2D4; optional date 2027-01-01.
2. Exact topic	Systemd security and privilege boundaries
3. Exact subtopics	Unit sandboxing, capabilities, namespaces, DynamicUser, ProtectSystem, NoNewPrivileges and cgroup limits
4. Learning objective	Explain how service privileges can be reduced without rewriting an application
5. Primary lecture / reading + direct URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	systemd.unit manual and systemd.exec security options; use current manual search for exact directives. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a benign unit that writes to one directory, add three sandbox directives, test failure/success, and explain each restriction. **2.** Standard (~2 h): Create a benign unit that writes to one directory, add three sandbox directives, test failure/success, and explain each restriction. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a benign unit that writes to one directory, add three sandbox directives, test failure/success, and explain each restriction. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	systemd-hardening.service and test log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a file write fail even when traditional Unix permissions appear correct? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Service works under the intended policy and failures are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "systemd.unit manual and systemd.exec security options; use current manual search for exact directives"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux man-pages project; write a five-line correction; retest next morning with a new 10-minute problem.

D068 · M03W2D5 · OverTheWire Bandit intermediate independent lab

Phase: Academic month · **Optional date:** 2027-01-02 · **Domain resources:** L02 / L03

1. Day / sequence	Global Day 68 of 365; calendar-free M03W2D5; optional date 2027-01-02.
2. Exact topic	OverTheWire Bandit intermediate independent lab
3. Exact subtopics	Pipes, redirection, permissions, hidden files, encoding, compression and shell escapes
4. Learning objective	Solve unfamiliar command-line puzzles by reading man pages rather than following a walkthrough
5. Primary lecture / reading + direct URL	OverTheWire Bandit `https://overthewire.org/wargames/bandit/` **Type:** CTF / command-line lab · **Access:** Free
6. Exact section / lesson locator	OverTheWire Bandit Levels 11-20; use official level descriptions and local man pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve at least six levels, document the concept and exact reasoning for each, and write one alternate command that would also work. **2.** Standard (~2 h): Solve at least six levels, document the concept and exact reasoning for each, and write one alternate command that would also work. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve at least six levels, document the concept and exact reasoning for each, and write one alternate command that would also work. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	bandit-intermediate.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What assumption did each level exploit, and how could you verify it locally? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Six levels solved without copied passwords or step-by-step tutorials Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OverTheWire Bandit Levels 11-20; use official level descriptions and local man pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux Journey; write a five-line correction; retest next morning with a new 10-minute problem.

D069 · M03W2D6 · Weekly Linux practical and troubleshooting

Phase: Academic month · **Optional date:** 2027-01-03 · **Domain resources:** L01 / L03

1. Day / sequence	Global Day 69 of 365; calendar-free M03W2D6; optional date 2027-01-03.
2. Exact topic	Weekly Linux practical and troubleshooting
3. Exact subtopics	Filesystem, users, processes, Bash, networking, logs, hardening, systemd and independent puzzles
4. Learning objective	Demonstrate Linux administration under a timed scenario
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey review sections selected from your error log. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 60-minute practical: create a user, harden a service, parse a log, fix one network issue and produce evidence. **2.** Standard (~2 h): Complete a 60-minute practical: create a user, harden a service, parse a log, fix one network issue and produce evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 60-minute practical: create a user, harden a service, parse a log, fix one network issue and produce evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3W2-practical.md and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would you check if a service starts manually but fails at boot? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and every failing step has a diagnostic note Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey review sections selected from your error log"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D070 · M03W2D7 · Windows architecture, accounts and permissions

Phase: Academic month · **Optional date:** 2027-01-04 · **Domain resources:** F14 / L07 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 70 of 365; calendar-free M03W2D7; optional date 2027-01-04.
2. Exact topic	Windows architecture, accounts and permissions
3. Exact subtopics	NTFS permissions, UAC, SID, access token, local accounts, groups, registry basics and process integrity
4. Learning objective	Compare Windows identity and access control with Linux equivalents
5. Primary lecture / reading + direct URL	Microsoft Learn Windows fundamentals `https://learn.microsoft.com/en-us/training/modules/explore-windows-client/` **Type:** Official learning hub · **Access:** Free learning content; exams separate
6. Exact section / lesson locator	Microsoft Learn, Explore the Windows client module and Windows architecture module. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Windows Server documentation `https://learn.microsoft.com/en-us/windows-server/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — In a Windows VM, inspect a local user, group, file ACL and process token; build a comparison table of Linux and Windows concepts — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — In a Windows VM, inspect a local user, group, file ACL and process token; build a comparison table of Linux and Windows concepts — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — In a Windows VM, inspect a local user, group, file ACL and process token; build a comparison table of Linux and Windows concepts — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-identity.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why is a SID more stable than a username for access control? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table is accurate and observations come from the lab VM Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Learn, Explore the Windows client module and Windows architecture module"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PowerShell documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 · Week 3 · daily cards

D071 · M03W3D1 · Windows processes, services and Sysinternals

Phase: Academic month · **Optional date:** 2027-01-05 · **Domain resources:** L10 / L08

1. Day / sequence	Global Day 71 of 365; calendar-free M03W3D1; optional date 2027-01-05.
2. Exact topic	Windows processes, services and Sysinternals
3. Exact subtopics	Process tree, handles, DLLs, services, scheduled tasks, autoruns and Process Explorer/Process Monitor
4. Learning objective	Investigate a benign process launch and identify its parent, modules and file activity
5. Primary lecture / reading + direct URL	Microsoft Sysinternals `https://learn.microsoft.com/en-us/sysinternals/` **Type:** Official tools/docs · **Access:** Free
6. Exact section / lesson locator	Microsoft Sysinternals documentation for Process Explorer, Process Monitor and Autoruns. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Launch notepad or another benign app, inspect process properties, capture a short ProcMon trace, and explain three event classes. **2.** Standard (~2 h): Launch notepad or another benign app, inspect process properties, capture a short ProcMon trace, and explain three event classes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Launch notepad or another benign app, inspect process properties, capture a short ProcMon trace, and explain three event classes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-process-trace.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How can a process with a legitimate name still be suspicious? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Process tree and trace are consistent and no unrelated personal data is captured Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Sysinternals documentation for Process Explorer, Process Monitor and Autoruns"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PowerShell documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D072 · M03W3D2 · PowerShell fundamentals and safe scripting

Phase: Academic month · **Optional date:** 2027-01-06 · **Domain resources:** L08 / L03

1. Day / sequence	Global Day 72 of 365; calendar-free M03W3D2; optional date 2027-01-06.
2. Exact topic	PowerShell fundamentals and safe scripting
3. Exact subtopics	Cmdlets, pipeline, objects versus text, Get-Help, variables, execution policy, remoting basics and encoding
4. Learning objective	Write a benign Windows inventory script using PowerShell objects
5. Primary lecture / reading + direct URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	PowerShell documentation, getting started and scripting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run Get-Process, Get-Service and Get-LocalUser; export structured JSON/CSV; write a script that handles empty results and logs errors. **2.** Standard (~2 h): Run Get-Process, Get-Service and Get-LocalUser; export structured JSON/CSV; write a script that handles empty results and logs errors. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run Get-Process, Get-Service and Get-LocalUser; export structured JSON/CSV; write a script that handles empty results and logs errors. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-inventory.ps1 and sample output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why are PowerShell pipeline objects different from Bash text streams? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Script works, output is structured, and errors are handled without suppressing them Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PowerShell documentation, getting started and scripting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Greg's Wiki BashGuide; write a five-line correction; retest next morning with a new 10-minute problem.

D073 · M03W3D3 · Windows Event Viewer and event IDs

Phase: Academic month · **Optional date:** 2027-01-07 · **Domain resources:** L09 / L11

1. Day / sequence	Global Day 73 of 365; calendar-free M03W3D3; optional date 2027-01-07.
2. Exact topic	Windows Event Viewer and event IDs
3. Exact subtopics	Channels, providers, levels, EventID, RecordId, Security log, System log, XPath queries and EVTX export
4. Learning objective	Find and interpret security-relevant Windows events from your own lab
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation, event records and queries. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sysmon documentation and download `https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate logon success/failure and service start events, locate them in Event Viewer, export a sanitized EVTX and record key fields. **2.** Standard (~2 h): Generate logon success/failure and service start events, locate them in Event Viewer, export a sanitized EVTX and record key fields. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate logon success/failure and service start events, locate them in Event Viewer, export a sanitized EVTX and record key fields. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-events.csv and sanitization note; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can the same EventID require different interpretation depending on logon type or subject? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Events are found, fields are labelled and private identifiers are sanitized Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation, event records and queries"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D074 · M03W3D4 · Sysmon installation and process telemetry

Phase: Academic month · **Optional date:** 2027-01-08 · **Domain resources:** L11 / L09

1. Day / sequence	Global Day 74 of 365; calendar-free M03W3D4; optional date 2027-01-08.
2. Exact topic	Sysmon installation and process telemetry
3. Exact subtopics	Sysmon event IDs, configuration, process creation, network connection, image loads and hashing in a disposable VM
4. Learning objective	Collect high-value endpoint telemetry and understand its privacy implications
5. Primary lecture / reading + direct URL	Sysmon documentation and download `https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon` **Type:** Official tool/doc · **Access:** Free
6. Exact section / lesson locator	Sysmon documentation and event ID schema. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Install Sysmon with a minimal official/sample config in a disposable VM, generate a benign process and network connection, and export two events. **2.** Standard (~2 h): Install Sysmon with a minimal official/sample config in a disposable VM, generate a benign process and network connection, and export two events. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Install Sysmon with a minimal official/sample config in a disposable VM, generate a benign process and network connection, and export two events. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sysmon-config.xml and event samples; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which Sysmon field would you use to distinguish a renamed executable from a legitimate one? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sysmon events appear and the answer includes image, hash, command line or parent process Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sysmon documentation and event ID schema"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Windows Event Log documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D075 · M03W3D5 · Active Directory architecture and identity boundaries

Phase: Academic month · **Optional date:** 2027-01-09 · **Domain resources:** L12 / L07

1. Day / sequence	Global Day 75 of 365; calendar-free M03W3D5; optional date 2027-01-09.
2. Exact topic	Active Directory architecture and identity boundaries
3. Exact subtopics	Forest, domain, OU, domain controller, DNS, LDAP, Kerberos, users, groups, GPO, trusts and tiering concept
4. Learning objective	Diagram an AD environment and explain authentication versus authorization boundaries
5. Primary lecture / reading + direct URL	Active Directory Domain Services overview `https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Active Directory Domain Services overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Windows Server documentation `https://learn.microsoft.com/en-us/windows-server/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a detailed diagram of a fictional university domain with OUs, groups, GPOs and trust boundaries; do not deploy AD if hardware is insufficient. **2.** Standard (~2 h): Build a detailed diagram of a fictional university domain with OUs, groups, GPOs and trust boundaries; do not deploy AD if hardware is insufficient. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a detailed diagram of a fictional university domain with OUs, groups, GPOs and trust boundaries; do not deploy AD if hardware is insufficient. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ad-architecture.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should administrative accounts not be used for ordinary email and web browsing? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram correctly separates identity, administration and resource boundaries Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Active Directory Domain Services overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D076 · M03W3D6 · Weekly Windows and AD review

Phase: Academic month · **Optional date:** 2027-01-10 · **Domain resources:** L08 / L12

1. Day / sequence	Global Day 76 of 365; calendar-free M03W3D6; optional date 2027-01-10.
2. Exact topic	Weekly Windows and AD review
3. Exact subtopics	Windows architecture, permissions, processes, PowerShell, events, Sysmon and AD; retrieval practice
4. Learning objective	Consolidate Month 3 Week 3 and prepare endpoint hardening
5. Primary lecture / reading + direct URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	PowerShell documentation, cmdlet reference and remoting overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Active Directory Domain Services overview `https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 20 retrieval questions, explain five Windows event fields from memory, and map one synthetic attack path to AD trust boundaries. **2.** Standard (~2 h): Complete 20 retrieval questions, explain five Windows event fields from memory, and map one synthetic attack path to AD trust boundaries. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 20 retrieval questions, explain five Windows event fields from memory, and map one synthetic attack path to AD trust boundaries. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3W3-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which single log source would you add first to improve endpoint visibility and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70 percent and AD diagram passes peer-style self-review Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PowerShell documentation, cmdlet reference and remoting overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Greg's Wiki BashGuide; write a five-line correction; retest next morning with a new 10-minute problem.

D077 · M03W3D7 · Group Policy and security baselines

Phase: Academic month · **Optional date:** 2027-01-11 · **Domain resources:** L13 / L14 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 77 of 365; calendar-free M03W3D7; optional date 2027-01-11.
2. Exact topic	Group Policy and security baselines
3. Exact subtopics	GPO processing, scope, precedence, security settings, audit policy, baseline packages and policy analyzer
4. Learning objective	Apply a safe lab GPO or, without a domain, interpret a baseline and predict its effect
5. Primary lecture / reading + direct URL	Windows security baselines `https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines` **Type:** Official configuration guidance · **Access:** Free
6. Exact section / lesson locator	Windows security baselines, deployment and analyzer guidance. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Benchmarks `https://www.cisecurity.org/cis-benchmarks`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a GPO plan for password policy, audit policy and service restriction; test one setting locally if no domain controller exists — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a GPO plan for password policy, audit policy and service restriction; test one setting locally if no domain controller exists — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a GPO plan for password policy, audit policy and service restriction; test one setting locally if no domain controller exists — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	gpo-plan.md and before/after screenshots; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why can a GPO setting fail to apply even though it is configured at the domain level? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes scope, precedence, validation and rollback Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows security baselines, deployment and analyzer guidance"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Benchmarks; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 · Week 4 · daily cards

D078 · M03W4D1 · Endpoint security controls and patch management

Phase: Academic month · **Optional date:** 2027-01-12 · **Domain resources:** C01 / C21

1. Day / sequence	Global Day 78 of 365; calendar-free M03W4D1; optional date 2027-01-12.
2. Exact topic	Endpoint security controls and patch management
3. Exact subtopics	Defender/antivirus concepts, EDR/telemetry, attack surface reduction, application control, patch rings and vulnerability remediation
4. Learning objective	Design a small endpoint security policy with measurable validation tests
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/ **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 objective 2 and 4 videos on endpoint security and mitigation techniques. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 https://www.cisecurity.org/controls/v8
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: On a Windows VM, inspect current protections and update state; create a hardening checklist with ten controls and tests without disabling essential protections. **2.** Standard (~2 h): On a Windows VM, inspect current protections and update state; create a hardening checklist with ten controls and tests without disabling essential protections. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): On a Windows VM, inspect current protections and update state; create a hardening checklist with ten controls and tests without disabling essential protections. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	endpoint-baseline.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is disabling security controls as a routine practice unacceptable even in a lab? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Checklist contains safe controls and no instruction to turn off host protection Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 objective 2 and 4 videos on endpoint security and mitigation techniques"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D079 · M03W4D2 · Windows image and event-log investigation project

Phase: Academic month · **Optional date:** 2027-01-13 · **Domain resources:** L09 / B05

1. Day / sequence	Global Day 79 of 365; calendar-free M03W4D2; optional date 2027-01-13.
2. Exact topic	Windows image and event-log investigation project
3. Exact subtopics	Evidence acquisition, timeline, logon events, service events, Sysmon process creation and report writing
4. Learning objective	Investigate a benign simulated incident from provided or self-generated logs
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation and B05 EVTX sample repository for optional extra samples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a timeline of five synthetic events, identify the root cause injected in the dataset, and write an incident report with confidence levels. **2.** Standard (~2 h): Create a timeline of five synthetic events, identify the root cause injected in the dataset, and write an incident report with confidence levels. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a timeline of five synthetic events, identify the root cause injected in the dataset, and write an incident report with confidence levels. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-06 evidence package and incident-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence corroborates your conclusion and what alternative explanation remains? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is consistent, report states limitations, and all data is sanitized Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation and B05 EVTX sample repository for optional extra samples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D080 · M03W4D3 · Linux and Windows audit utility project

Phase: Academic month · **Optional date:** 2027-01-14 · **Domain resources:** L03 / P05

1. Day / sequence	Global Day 80 of 365; calendar-free M03W4D3; optional date 2027-01-14.
2. Exact topic	Linux and Windows audit utility project
3. Exact subtopics	Inventory, permission checks, service state, patch state, listening ports, log collection and JSON output
4. Learning objective	Build a useful defensive automation tool that works in the student's own lab
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	BashGuide functions, tests and quoting sections; L08 for PowerShell objects. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write separate Bash and PowerShell scripts that output a common JSON schema; add tests for empty values and malformed input. **2.** Standard (~2 h): Write separate Bash and PowerShell scripts that output a common JSON schema; add tests for empty values and malformed input. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write separate Bash and PowerShell scripts that output a common JSON schema; add tests for empty values and malformed input. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	audit-tool repository with README and test results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must an audit tool avoid executing untrusted file contents? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Both scripts run, tests pass, and no secrets or personal data are in sample output Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "BashGuide functions, tests and quoting sections; L08 for PowerShell objects"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D081 · M03W4D4 · Month 3 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-01-15 · **Domain resources:** L01 / L09

1. Day / sequence	Global Day 81 of 365; calendar-free M03W4D4; optional date 2027-01-15.
2. Exact topic	Month 3 practical examination and milestone gate
3. Exact subtopics	Linux administration, Windows administration, PowerShell, logs, AD architecture, hardening and scripting
4. Learning objective	Demonstrate system-administration competence with independent troubleshooting
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey review sections chosen from the error log. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 120-minute practical: harden Linux and Windows artifacts, interpret logs, write a PowerShell query and explain an AD diagram. **2.** Standard (~2 h): Complete a 120-minute practical: harden Linux and Windows artifacts, interpret logs, write a PowerShell query and explain an AD diagram. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 120-minute practical: harden Linux and Windows artifacts, interpret logs, write a PowerShell query and explain an AD diagram. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you determine whether an account change was authorized using only available logs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey review sections chosen from the error log"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D082 · M03W4D5 · Monthly review, cleanup and Month 4 preview

Phase: Academic month · **Optional date:** 2027-01-16 · **Domain resources:** L03 / F08

1. Day / sequence	Global Day 82 of 365; calendar-free M03W4D5; optional date 2027-01-16.
2. Exact topic	Monthly review, cleanup and Month 4 preview
3. Exact subtopics	Command notebook, error log, script bugs, AD misconceptions and project limitations
4. Learning objective	Repair Month 3 gaps and prepare programming automation prerequisites
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	BashGuide review sections and P04 RegexOne review lessons. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Revisit failed assessment items, refactor two scripts, produce ten flashcards, and install Python/testing tools in a clean virtual environment. **2.** Standard (~2 h): Revisit failed assessment items, refactor two scripts, produce ten flashcards, and install Python/testing tools in a clean virtual environment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Revisit failed assessment items, refactor two scripts, produce ten flashcards, and install Python/testing tools in a clean virtual environment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3-review.md and tools-check.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which programming concept is the biggest bottleneck for your security automation goal? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest passes, lab snapshots are clean, and Python environment is ready Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "BashGuide review sections and P04 RegexOne review lessons"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D083 · M03W4D6 · Kerberos, NTLM and AD authentication flows

Phase: Academic month · **Optional date:** 2027-01-17 · **Domain resources:** L12 / C18

1. Day / sequence	Global Day 83 of 365; calendar-free M03W4D6; optional date 2027-01-17.
2. Exact topic	Kerberos, NTLM and AD authentication flows
3. Exact subtopics	Ticket-granting ticket, service ticket, KDC, realm, SPN, NTLM challenge-response, golden/silver ticket concept and pass-the-hash concept
4. Learning objective	Explain how AD authentication differs from simply sending a password to each service
5. Primary lecture / reading + direct URL	Active Directory Domain Services overview `https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Active Directory Domain Services overview and Microsoft authentication documentation linked from that page. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a Kerberos login and service-access sequence; identify where a credential or ticket could be stolen and one detection data source. **2.** Standard (~2 h): Draw a Kerberos login and service-access sequence; identify where a credential or ticket could be stolen and one detection data source. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a Kerberos login and service-access sequence; identify where a credential or ticket could be stolen and one detection data source. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	kerberos-flow.svg and ATT&CK mapping; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a stolen ticket useful even if the attacker does not know the plaintext password? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sequence is correct and detection data sources are specific rather than tool-name lists Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Active Directory Domain Services overview and Microsoft authentication documentation linked from that page"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D084 · M03W4D7 · Spaced revision, recovery and portfolio update

Phase: Academic month · **Optional date:** 2027-01-18 · **Domain resources:** F09 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 84 of 365; calendar-free M03W4D7; optional date 2027-01-18.
2. Exact topic	Spaced revision, recovery and portfolio update
3. Exact subtopics	Linux and Windows command retrieval, event IDs, permissions and identity boundaries
4. Learning objective	Preserve Month 3 gains and publish sanitized project evidence
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapter 2 for clean commits and tags. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Update PRJ-05 and PRJ-06 READMEs, add limitations and validation tests, tag releases, and do a light 30-minute retrieval session — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Update PRJ-05 and PRJ-06 READMEs, add limitations and validation tests, tag releases, and do a light 30-minute retrieval session — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Update PRJ-05 and PRJ-06 READMEs, add limitations and validation tests, tag releases, and do a light 30-minute retrieval session — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	updated portfolio links and M3-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which three commands or event fields do you still confuse, and when will you review them? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repositories contain no secrets and each project includes reproducible instructions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapter 2 for clean commits and tags"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 — Python, scripting, Git and security automation

Days D085-D112. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Python setup, execution model and development workflow D2: Variables, primitive data types and expressions D3: Collections: lists, tuples, sets and dictionaries D4: Control flow, conditions and loops D5: Functions, scope, parameters and return values D6: Files, paths, exceptions and context managers D7: Weekly Python fundamentals review and mini project
Week 2	D1: Regular expressions and defensive parsing D2: Modules, packages, imports and virtual environments D3: CSV, JSON and data normalization D4: HTTP requests, APIs and safe client behavior D5: SQL and parameterized database access D6: Logging, observability and defensive error handling D7: Weekly automation review and debugging practical
Week 3	D1: Bash for security automation D2: PowerShell objects and structured output D3: Git workflows, code review and documentation D4: Testing security utilities and adversarial inputs D5: Weekly automation practical and code quality gate D6: Security audit utility design and requirements D7: Log parser engineering sprint: parser and normalization
Week 4	D1: Log parser engineering sprint: analysis and reporting D2: Secure coding review of the automation tool D3: Bash and PowerShell automation comparison lab D4: Project publication and professional README D5: Month 4 practical assessment and milestone review D6: Monthly review and prerequisite gate for security principles D7: Spaced revision, recovery and Month 5 preview

> **Gate:** score the Month 4 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 04 · Week 1 · daily cards

D085 · M04W1D1 · Python setup, execution model and development workflow

Phase: Academic month · **Optional date:** 2027-01-19 · **Domain resources:** F08 / P01

1. Day / sequence	Global Day 85 of 365; calendar-free M04W1D1; optional date 2027-01-19.
2. Exact topic	Python setup, execution model and development workflow
3. Exact subtopics	Interpreter, REPL, scripts, virtual environments, pip, modules, indentation and syntax errors
4. Learning objective	Set up a clean Python environment and explain how a script runs
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 1-2: Whetting Your Appetite and Using the Python Interpreter. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python for Everybody `https://www.py4e.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a venv, install one harmless library, run three scripts that fail in different ways, and document the tracebacks. **2.** Standard (~2 h): Create a venv, install one harmless library, run three scripts that fail in different ways, and document the tracebacks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a venv, install one harmless library, run three scripts that fail in different ways, and document the tracebacks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	python-setup.md and error-log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What does a traceback tell you about the call stack and the first real error? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Virtual environment is reproducible and all three errors are classified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 1-2: Whetting Your Appetite and Using the Python Interpreter"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D086 · M04W1D2 · Variables, primitive data types and expressions

Phase: Academic month · **Optional date:** 2027-01-20 · **Domain resources:** F08 / P01

1. Day / sequence	Global Day 86 of 365; calendar-free M04W1D2; optional date 2027-01-20.
2. Exact topic	Variables, primitive data types and expressions
3. Exact subtopics	Integers, floats, strings, booleans, None, operators, type conversion, immutability and f-strings
4. Learning objective	Predict types and values of expressions without running code
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 3: An Informal Introduction to Python. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python for Everybody `https://www.py4e.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 30 expression-prediction exercises; verify with the interpreter and correct wrong predictions in the error log. **2.** Standard (~2 h): Complete 30 expression-prediction exercises; verify with the interpreter and correct wrong predictions in the error log. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 30 expression-prediction exercises; verify with the interpreter and correct wrong predictions in the error log. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	expressions.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does string concatenation with an integer raise TypeError and how can that prevent silent data corruption? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least 85 percent predictions correct and all mistakes explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 3: An Informal Introduction to Python"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D087 · M04W1D3 · Collections: lists, tuples, sets and dictionaries

Phase: Academic month · **Optional date:** 2027-01-21 · **Domain resources:** F08 / P02

1. Day / sequence	Global Day 87 of 365; calendar-free M04W1D3; optional date 2027-01-21.
2. Exact topic	Collections: lists, tuples, sets and dictionaries
3. Exact subtopics	Indexing, slicing, mutability, comprehensions, dictionaries, sets, nested structures and copying
4. Learning objective	Choose the appropriate collection for a security-data task
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 3-5: informal introduction, control flow and data structures. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Learn Python - Full Course for Beginners (freeCodeCamp) `https://www.youtube.com/watch?v=eWRfhZUzrAc`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Store ten synthetic events in each collection type, write queries to count unique IPs, and explain shallow versus deep copies. **2.** Standard (~2 h): Store ten synthetic events in each collection type, write queries to count unique IPs, and explain shallow versus deep copies. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Store ten synthetic events in each collection type, write queries to count unique IPs, and explain shallow versus deep copies. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	collections-lab.py; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: When is a set better than a list for IOC lookup, and what information is lost? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All programs run and the collection choices are justified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 3-5: informal introduction, control flow and data structures"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D088 · M04W1D4 · Control flow, conditions and loops

Phase: Academic month · **Optional date:** 2027-01-22 · **Domain resources:** F08 / P02

1. Day / sequence	Global Day 88 of 365; calendar-free M04W1D4; optional date 2027-01-22.
2. Exact topic	Control flow, conditions and loops
3. Exact subtopics	Boolean logic, comparison, if/elif/else, for/while, break/continue, range, enumerate, zip and off-by-one errors
4. Learning objective	Implement a robust loop with explicit termination and edge-case handling
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 4: More Control Flow Tools. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Learn Python - Full Course for Beginners (freeCodeCamp) `https://www.youtube.com/watch?v=eWRfhZUzrAc`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a parser that classifies 100 synthetic log lines by severity; include empty lines, unknown severity and very long lines. **2.** Standard (~2 h): Write a parser that classifies 100 synthetic log lines by severity; include empty lines, unknown severity and very long lines. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a parser that classifies 100 synthetic log lines by severity; include empty lines, unknown severity and very long lines. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	classifier.py and test-results.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which loop invariant guarantees that every line is processed exactly once? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All edge cases are handled and output counts match a manually verified sample Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 4: More Control Flow Tools"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D089 · M04W1D5 · Functions, scope, parameters and return values

Phase: Academic month · **Optional date:** 2027-01-23 · **Domain resources:** F08 / P05

1. Day / sequence	Global Day 89 of 365; calendar-free M04W1D5; optional date 2027-01-23.
2. Exact topic	Functions, scope, parameters and return values
3. Exact subtopics	Function definitions, arguments, defaults, keyword arguments, return values, local/global scope and docstrings
4. Learning objective	Design reusable functions with clear contracts and no hidden side effects
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 4.6-4.8: defining functions, default arguments and coding style. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Refactor the classifier into five small functions; write docstrings and unit tests for normal, malformed and boundary inputs. **2.** Standard (~2 h): Refactor the classifier into five small functions; write docstrings and unit tests for normal, malformed and boundary inputs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Refactor the classifier into five small functions; write docstrings and unit tests for normal, malformed and boundary inputs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	functions.py and pytest results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a mutable default argument cause surprising state persistence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests pass and no hidden global state remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 4.6-4.8: defining functions, default arguments and coding style"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D090 · M04W1D6 · Files, paths, exceptions and context managers

Phase: Academic month · **Optional date:** 2027-01-24 · **Domain resources:** F08 / P03

1. Day / sequence	Global Day 90 of 365; calendar-free M04W1D6; optional date 2027-01-24.
2. Exact topic	Files, paths, exceptions and context managers
3. Exact subtopics	open modes, pathlib, with-statement, encoding, CSV, JSON, exceptions, try/except/else/finally and atomic writes
4. Learning objective	Read and write structured security data without corrupting files
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 7 and 8: input/output and errors/exceptions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Convert a synthetic auth.log to JSON and CSV, force a UnicodeDecodeError and a JSONDecodeError, then handle both safely. **2.** Standard (~2 h): Convert a synthetic auth.log to JSON and CSV, force a UnicodeDecodeError and a JSONDecodeError, then handle both safely. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Convert a synthetic auth.log to JSON and CSV, force a UnicodeDecodeError and a JSONDecodeError, then handle both safely. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	file-conversion.py and error tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should you write to a temporary file and rename it atomically for important outputs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Conversion works, exceptions are explicit, and no partial output is accepted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 7 and 8: input/output and errors/exceptions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D091 · M04W1D7 · Weekly Python fundamentals review and mini project

Phase: Academic month · **Optional date:** 2027-01-25 · **Domain resources:** F08 / P05 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 91 of 365; calendar-free M04W1D7; optional date 2027-01-25.
2. Exact topic	Weekly Python fundamentals review and mini project
3. Exact subtopics	Types, collections, conditions, loops, functions, files, exceptions and debugging
4. Learning objective	Build a small independent tool from specifications rather than a tutorial
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 1-8 review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a CLI tool that reads a synthetic log and prints a summary; include tests, a README and one deliberate bug fixed through debugging — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a CLI tool that reads a synthetic log and prints a summary; include tests, a README and one deliberate bug fixed through debugging — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a CLI tool that reads a synthetic log and prints a summary; include tests, a README and one deliberate bug fixed through debugging — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	mini-log-summary repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which test caught the most important bug and what did it assume? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 on the monthly rubric and all tests pass Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 1-8 review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 · Week 2 · daily cards

D092 · M04W2D1 · Regular expressions and defensive parsing

Phase: Academic month · **Optional date:** 2027-01-26 · **Domain resources:** P04 / F08

1. Day / sequence	Global Day 92 of 365; calendar-free M04W2D1; optional date 2027-01-26.
2. Exact topic	Regular expressions and defensive parsing
3. Exact subtopics	Character classes, quantifiers, anchors, groups, named groups, lookarounds, greedy/lazy matching and catastrophic backtracking risk
4. Learning objective	Parse untrusted logs with bounded, readable patterns and avoid regex denial of service
5. Primary lecture / reading + direct URL	RegexOne `https://regexone.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	RegexOne Lessons 1-15, especially practical examples and backreferences. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write regexes for Apache, SSH and JSON-ish log lines; test against 50 valid and 10 malformed lines; measure one pathological pattern safely. **2.** Standard (~2 h): Write regexes for Apache, SSH and JSON-ish log lines; test against 50 valid and 10 malformed lines; measure one pathological pattern safely. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write regexes for Apache, SSH and JSON-ish log lines; test against 50 valid and 10 malformed lines; measure one pathological pattern safely. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	regex-lab.py and timing notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should a regex for untrusted input avoid nested ambiguous quantifiers? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All valid lines parse, malformed lines fail safely, and regex limitations are documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "RegexOne Lessons 1-15, especially practical examples and backreferences"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D093 · M04W2D2 · Modules, packages, imports and virtual environments

Phase: Academic month · **Optional date:** 2027-01-27 · **Domain resources:** F08 / W13

1. Day / sequence	Global Day 93 of 365; calendar-free M04W2D2; optional date 2027-01-27.
2. Exact topic	Modules, packages, imports and virtual environments
3. Exact subtopics	Module search path, __name__, packages, pip, requirements.txt, pyproject.toml, dependency pinning and malicious package risk
4. Learning objective	Structure a small project and explain supply-chain trust boundaries
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 6: Modules. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Split the log summary into parser, analyzer and CLI modules; pin dependencies; inspect package metadata and write a dependency-risk note. **2.** Standard (~2 h): Split the log summary into parser, analyzer and CLI modules; pin dependencies; inspect package metadata and write a dependency-risk note. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Split the log summary into parser, analyzer and CLI modules; pin dependencies; inspect package metadata and write a dependency-risk note. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	project-structure.md and requirements.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a package name typo a security issue rather than just a spelling mistake? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Package imports work offline and dependency decisions are documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 6: Modules"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D094 · M04W2D3 · CSV, JSON and data normalization

Phase: Academic month · **Optional date:** 2027-01-28 · **Domain resources:** P03 / P07

1. Day / sequence	Global Day 94 of 365; calendar-free M04W2D3; optional date 2027-01-28.
2. Exact topic	CSV, JSON and data normalization
3. Exact subtopics	csv module, json module, schemas, nested JSON, data types, timestamp formats, missing fields and canonicalization
4. Learning objective	Normalize three log formats into one common event schema
5. Primary lecture / reading + direct URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/` **Type:** Free online book · **Access:** Free online; paid print/course optional
6. Exact section / lesson locator	Automate the Boring Stuff, CSV/JSON chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SQLite documentation `https://www.sqlite.org/docs.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create three synthetic input files with inconsistent fields, normalize them into common JSON, and reject records without required fields. **2.** Standard (~2 h): Create three synthetic input files with inconsistent fields, normalize them into common JSON, and reject records without required fields. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create three synthetic input files with inconsistent fields, normalize them into common JSON, and reject records without required fields. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	normalized-events.json and schema.json; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must timestamps include a time zone before correlating events? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every output record satisfies the schema and all rejected records have a reason Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Automate the Boring Stuff, CSV/JSON chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D095 · M04W2D4 · HTTP requests, APIs and safe client behavior

Phase: Academic month · **Optional date:** 2027-01-29 · **Domain resources:** P06 / F10

1. Day / sequence	Global Day 95 of 365; calendar-free M04W2D4; optional date 2027-01-29.
2. Exact topic	HTTP requests, APIs and safe client behavior
3. Exact subtopics	requests, GET/POST, status codes, headers, query parameters, timeouts, retries, TLS verification, sessions and rate limits
4. Learning objective	Query an API safely and handle network failures without disabling security controls
5. Primary lecture / reading + direct URL	Requests: HTTP for Humans `https://requests.readthedocs.io/en/latest/` **Type:** Official library docs · **Access:** Free
6. Exact section / lesson locator	Requests Quickstart and Advanced Usage, especially timeouts, SSL verification and sessions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Send requests to a local mock API; test timeout, 4xx, 5xx, malformed JSON and rate-limit response; implement bounded retry logic. **2.** Standard (~2 h): Send requests to a local mock API; test timeout, 4xx, 5xx, malformed JSON and rate-limit response; implement bounded retry logic. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Send requests to a local mock API; test timeout, 4xx, 5xx, malformed JSON and rate-limit response; implement bounded retry logic. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	api-client.py and test log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is verify=False dangerous and what is the correct local-lab alternative? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All failure modes are handled and TLS verification remains enabled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Requests Quickstart and Advanced Usage, especially timeouts, SSL verification and sessions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP Overview; write a five-line correction; retest next morning with a new 10-minute problem.

D096 · M04W2D5 · SQL and parameterized database access

Phase: Academic month · **Optional date:** 2027-01-30 · **Domain resources:** P07 / F11

1. Day / sequence	Global Day 96 of 365; calendar-free M04W2D5; optional date 2027-01-30.
2. Exact topic	SQL and parameterized database access
3. Exact subtopics	sqlite3, parameterized queries, transactions, indexes, joins, GROUP BY, injection risk and least-privilege database accounts
4. Learning objective	Store and query security events safely with parameterized SQL
5. Primary lecture / reading + direct URL	SQLite documentation `https://www.sqlite.org/docs.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	SQLite documentation, SQL syntax, transactions and security considerations. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SQLBolt `https://sqlbolt.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a database of normalized events, write five analytical queries, and demonstrate that a malicious string cannot alter query structure. **2.** Standard (~2 h): Build a database of normalized events, write five analytical queries, and demonstrate that a malicious string cannot alter query structure. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a database of normalized events, write five analytical queries, and demonstrate that a malicious string cannot alter query structure. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	events.db, queries.sql and injection-test.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does parameterization preserve the SQL parse tree while string concatenation does not? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Queries are correct and injection tests are safely performed on synthetic data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLite documentation, SQL syntax, transactions and security considerations"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SQLBolt; write a five-line correction; retest next morning with a new 10-minute problem.

D097 · M04W2D6 · Logging, observability and defensive error handling

Phase: Academic month · **Optional date:** 2027-01-31 · **Domain resources:** F08 / W17

1. Day / sequence	Global Day 97 of 365; calendar-free M04W2D6; optional date 2027-01-31.
2. Exact topic	Logging, observability and defensive error handling
3. Exact subtopics	logging module, levels, structured logs, correlation IDs, exception logging, secret redaction and avoiding noisy logs
4. Learning objective	Instrument a security utility so failures are diagnosable without leaking secrets
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Standard Library logging HOWTO linked from Python Tutorial Section 8. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add structured logging and redaction to the log parser; generate debug/info/warning/error events and verify no sample secrets in logs. **2.** Standard (~2 h): Add structured logging and redaction to the log parser; generate debug/info/warning/error events and verify no sample secrets in logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add structured logging and redaction to the log parser; generate debug/info/warning/error events and verify no sample secrets in logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	structured-logs.jsonl; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which fields are useful for incident response but unsafe to log without redaction? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Logs are useful for debugging and all synthetic secrets are redacted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Standard Library logging HOWTO linked from Python Tutorial Section 8"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D098 · M04W2D7 · Weekly automation review and debugging practical

Phase: Academic month · **Optional date:** 2027-02-01 · **Domain resources:** P05 / P03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 98 of 365; calendar-free M04W2D7; optional date 2027-02-01.
2. Exact topic	Weekly automation review and debugging practical
3. Exact subtopics	Modules, files, regex, APIs, SQL, logging and error handling; independent debugging
4. Learning objective	Repair an unfamiliar broken script using a disciplined debugging workflow
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation, getting started and assertion rewriting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Instructor/self-created broken parser with five bugs; reproduce, isolate, fix, add regression tests, and write a postmortem — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Instructor/self-created broken parser with five bugs; reproduce, isolate, fix, add regression tests, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Instructor/self-created broken parser with five bugs; reproduce, isolate, fix, add regression tests, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	debugging-postmortem.md and tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which two pieces of evidence narrowed each bug to a single function? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All five bugs are fixed and regression tests pass Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation, getting started and assertion rewriting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 · Week 3 · daily cards

D099 · M04W3D1 · Bash for security automation

Phase: Academic month · **Optional date:** 2027-02-02 · **Domain resources:** L03 / P03

1. Day / sequence	Global Day 99 of 365; calendar-free M04W3D1; optional date 2027-02-02.
2. Exact topic	Bash for security automation
3. Exact subtopics	set -euo pipefail, functions, arrays, traps, getopts, quoting, exit codes, process substitution and safe temporary files
4. Learning objective	Write a robust CLI audit wrapper around the Python tool
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	Greg's Wiki BashGuide, quoting, arrays, functions and traps sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a Bash wrapper that validates arguments, runs the parser in the venv, handles signals and cleans temporary files. **2.** Standard (~2 h): Create a Bash wrapper that validates arguments, runs the parser in the venv, handles signals and cleans temporary files. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a Bash wrapper that validates arguments, runs the parser in the venv, handles signals and cleans temporary files. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	run-audit.sh and test cases; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can an unquoted glob expression delete unintended files? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Wrapper passes normal, empty and malformed argument tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Greg's Wiki BashGuide, quoting, arrays, functions and traps sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D100 · M04W3D2 · PowerShell objects and structured output

Phase: Academic month · **Optional date:** 2027-02-03 · **Domain resources:** L08 / F09

1. Day / sequence	Global Day 100 of 365; calendar-free M04W3D2; optional date 2027-02-03.
2. Exact topic	PowerShell objects and structured output
3. Exact subtopics	Cmdlets, objects, formatting, ConvertTo-Json, error handling, execution policy and safe invocation
4. Learning objective	Build a Windows counterpart and compare its output schema with Bash/Python
5. Primary lecture / reading + direct URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	PowerShell documentation, scripting, modules and error handling sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a PowerShell inventory script that outputs the same common JSON schema as the Linux audit tool; test on a Windows VM. **2.** Standard (~2 h): Write a PowerShell inventory script that outputs the same common JSON schema as the Linux audit tool; test on a Windows VM. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a PowerShell inventory script that outputs the same common JSON schema as the Linux audit tool; test on a Windows VM. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-inventory.ps1 and sample JSON; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is Export-Csv with objects safer than parsing formatted table text? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Output parses as JSON and field names match the shared schema Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PowerShell documentation, scripting, modules and error handling sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Greg's Wiki BashGuide; write a five-line correction; retest next morning with a new 10-minute problem.

D101 · M04W3D3 · Git workflows, code review and documentation

Phase: Academic month · **Optional date:** 2027-02-04 · **Domain resources:** F09 / W13

1. Day / sequence	Global Day 101 of 365; calendar-free M04W3D3; optional date 2027-02-04.
2. Exact topic	Git workflows, code review and documentation
3. Exact subtopics	Feature branches, diff review, semantic commit messages, issues, pull requests, tags, CHANGELOG and reproducible setup
4. Learning objective	Use version control as professional engineering evidence rather than a file dump
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and Chapter 5 distributed Git basics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a feature branch for the parser, write a pull request to yourself, review the diff, fix two issues, tag v0.1 and write a CHANGELOG. **2.** Standard (~2 h): Create a feature branch for the parser, write a pull request to yourself, review the diff, fix two issues, tag v0.1 and write a CHANGELOG. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a feature branch for the parser, write a pull request to yourself, review the diff, fix two issues, tag v0.1 and write a CHANGELOG. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PR screenshot/diff and CHANGELOG.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence in a commit history shows independent engineering rather than copied code? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Clean history, meaningful commits and reproducible setup instructions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and Chapter 5 distributed Git basics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D102 · M04W3D4 · Testing security utilities and adversarial inputs

Phase: Academic month · **Optional date:** 2027-02-05 · **Domain resources:** P05 / P04

1. Day / sequence	Global Day 102 of 365; calendar-free M04W3D4; optional date 2027-02-05.
2. Exact topic	Testing security utilities and adversarial inputs
3. Exact subtopics	Unit, integration and property-style tests; fixtures; parametrization; malformed inputs; boundary values; fuzzing concepts and failure injection
4. Learning objective	Prove that a defensive tool fails safely on hostile input
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation, fixtures, parametrization and marks sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RegexOne `https://regexone.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add tests for empty files, huge lines, Unicode, null bytes, invalid JSON, regex backtracking and missing permissions; use a simple random input generator. **2.** Standard (~2 h): Add tests for empty files, huge lines, Unicode, null bytes, invalid JSON, regex backtracking and missing permissions; use a simple random input generator. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add tests for empty files, huge lines, Unicode, null bytes, invalid JSON, regex backtracking and missing permissions; use a simple random input generator. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	test-results.xml and fuzz-corpus/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which input caused the worst runtime and how did you bound it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass, runtime is bounded, and no crashes expose tracebacks to users Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation, fixtures, parametrization and marks sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D103 · M04W3D5 · Weekly automation practical and code quality gate

Phase: Academic month · **Optional date:** 2027-02-06 · **Domain resources:** P05 / F09

1. Day / sequence	Global Day 103 of 365; calendar-free M04W3D5; optional date 2027-02-06.
2. Exact topic	Weekly automation practical and code quality gate
3. Exact subtopics	Python, Bash, PowerShell, Git, APIs, SQL, regex, testing and documentation
4. Learning objective	Deliver a small independent automation tool with tests and code review
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and P03 Automate the Boring Stuff project chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a CLI that ingests two log formats, normalizes them, stores events, queries top talkers and outputs JSON/CSV; include a README and threat model. **2.** Standard (~2 h): Build a CLI that ingests two log formats, normalizes them, stores events, queries top talkers and outputs JSON/CSV; include a README and threat model. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a CLI that ingests two log formats, normalizes them, stores events, queries top talkers and outputs JSON/CSV; include a README and threat model. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	automation-tool repository and code-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Where could an attacker influence input, and how does your design constrain it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, tests pass, and documentation is sufficient for another student to run it Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and P03 Automate the Boring Stuff project chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D104 · M04W3D6 · Security audit utility design and requirements

Phase: Academic month · **Optional date:** 2027-02-07 · **Domain resources:** P08 / C21

1. Day / sequence	Global Day 104 of 365; calendar-free M04W3D6; optional date 2027-02-07.
2. Exact topic	Security audit utility design and requirements
3. Exact subtopics	Asset inventory, secure defaults, consent, scope, data minimization, read-only operations, exit codes and reporting
4. Learning objective	Design a safe, useful audit tool before adding features
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices Quick Reference Guide, input validation, error handling and data protection sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 `https://www.cisecurity.org/controls/v8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a one-page requirements and threat model for a Linux/Windows audit utility; define what it must never do. **2.** Standard (~2 h): Write a one-page requirements and threat model for a Linux/Windows audit utility; define what it must never do. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a one-page requirements and threat model for a Linux/Windows audit utility; define what it must never do. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	audit-tool-requirements.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which feature would create unacceptable risk and how would you redesign it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Requirements are testable and explicitly forbid credential collection and destructive changes Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices Quick Reference Guide, input validation, error handling and data protection sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D105 · M04W3D7 · Log parser engineering sprint: parser and normalization

Phase: Academic month · **Optional date:** 2027-02-08 · **Domain resources:** P03 / P05 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 105 of 365; calendar-free M04W3D7; optional date 2027-02-08.
2. Exact topic	Log parser engineering sprint: parser and normalization
3. Exact subtopics	Streaming input, multiple formats, schema validation, timestamp parsing, field extraction and error budgets
4. Learning objective	Implement an efficient parser that never crashes on malformed data
5. Primary lecture / reading + direct URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/` **Type:** Free online book · **Access:** Free online; paid print/course optional
6. Exact section / lesson locator	Automate the Boring Stuff chapters on regular expressions, files and debugging. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement streaming parsing for Apache, SSH and JSON lines; cap line length and errors; write 20 unit tests — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement streaming parsing for Apache, SSH and JSON lines; cap line length and errors; write 20 unit tests — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement streaming parsing for Apache, SSH and JSON lines; cap line length and errors; write 20 unit tests — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	parser.py and test_parser.py; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: How does streaming change memory usage and error recovery? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass and a 100 MB synthetic file can be processed within a documented memory limit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Automate the Boring Stuff chapters on regular expressions, files and debugging"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 · Week 4 · daily cards

D106 · M04W4D1 · Log parser engineering sprint: analysis and reporting

Phase: Academic month · **Optional date:** 2027-02-09 · **Domain resources:** P07 / P06

1. Day / sequence	Global Day 106 of 365; calendar-free M04W4D1; optional date 2027-02-09.
2. Exact topic	Log parser engineering sprint: analysis and reporting
3. Exact subtopics	Counting, top-N, time bucketing, correlation, output formats, plotting optional and statistical false positives
4. Learning objective	Generate a report that supports a conclusion rather than a data dump
5. Primary lecture / reading + direct URL	SQLite documentation `https://www.sqlite.org/docs.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	SQLite aggregation and indexing documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Requests: HTTP for Humans `https://requests.readthedocs.io/en/latest/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Query normalized events for failed logins by source, hour and username; create a Markdown report with confidence and limitations. **2.** Standard (~2 h): Query normalized events for failed logins by source, hour and username; create a Markdown report with confidence and limitations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Query normalized events for failed logins by source, hour and username; create a Markdown report with confidence and limitations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	security-log-report.md and SQL queries; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a high failed-login count be normal behaviour in some environments? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report includes baseline, context, limitations and at least one alternative explanation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLite aggregation and indexing documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SQLBolt; write a five-line correction; retest next morning with a new 10-minute problem.

D107 · M04W4D2 · Secure coding review of the automation tool

Phase: Academic month · **Optional date:** 2027-02-10 · **Domain resources:** P08 / W14

1. Day / sequence	Global Day 107 of 365; calendar-free M04W4D2; optional date 2027-02-10.
2. Exact topic	Secure coding review of the automation tool
3. Exact subtopics	Input validation, command injection, path traversal, secrets, dependency risk, logging, permissions and least privilege
4. Learning objective	Find and fix security flaws in your own code
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices and OWASP Cheat Sheet Series via W17. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Semgrep documentation `https://semgrep.dev/docs/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run Semgrep locally on the tool, triage findings, fix three real issues, and document one false positive. **2.** Standard (~2 h): Run Semgrep locally on the tool, triage findings, fix three real issues, and document one false positive. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run Semgrep locally on the tool, triage findings, fix three real issues, and document one false positive. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	secure-code-review.md and diff; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which bug would have allowed user-controlled input to reach a shell command? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Fixes are tested and no dangerous shell interpolation remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices and OWASP Cheat Sheet Series via W17"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D108 · M04W4D3 · Bash and PowerShell automation comparison lab

Phase: Academic month · **Optional date:** 2027-02-11 · **Domain resources:** L03 / P05

1. Day / sequence	Global Day 108 of 365; calendar-free M04W4D3; optional date 2027-02-11.
2. Exact topic	Bash and PowerShell automation comparison lab
3. Exact subtopics	Cross-platform telemetry schema, process inventory, listening ports, users, services and error handling
4. Learning objective	Produce equivalent evidence from Linux and Windows without leaking secrets
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	BashGuide review of arrays and quoting; L08 PowerShell error handling docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run both audit scripts, compare schemas, normalize differences, and explain two platform-specific limitations. **2.** Standard (~2 h): Run both audit scripts, compare schemas, normalize differences, and explain two platform-specific limitations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run both audit scripts, compare schemas, normalize differences, and explain two platform-specific limitations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cross-platform-report.md and sample outputs; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should both tools expose the same field names even when the underlying APIs differ? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Both tools produce valid common-schema JSON and limitations are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "BashGuide review of arrays and quoting; L08 PowerShell error handling docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D109 · M04W4D4 · Project publication and professional README

Phase: Academic month · **Optional date:** 2027-02-12 · **Domain resources:** F09 / W13

1. Day / sequence	Global Day 109 of 365; calendar-free M04W4D4; optional date 2027-02-12.
2. Exact topic	Project publication and professional README
3. Exact subtopics	Repository structure, setup, usage, examples, tests, screenshots, threat model, limitations, licence and changelog
4. Learning objective	Publish the security-log-parser and audit utility as portfolio evidence
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 1-3 and GitHub Docs README conventions linked from F09 repository navigation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Finalize PRJ-07, sanitize sample logs, add verification commands, tag a release and write a succinct professional README. **2.** Standard (~2 h): Finalize PRJ-07, sanitize sample logs, add verification commands, tag a release and write a succinct professional README. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Finalize PRJ-07, sanitize sample logs, add verification commands, tag a release and write a succinct professional README. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published PRJ-07 repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which three commands prove the project works on a clean machine? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository passes the portfolio checklist and contains no credentials or personal data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 1-3 and GitHub Docs README conventions linked from F09 repository navigation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D110 · M04W4D5 · Month 4 practical assessment and milestone review

Phase: Academic month · **Optional date:** 2027-02-13 · **Domain resources:** F08 / P05

1. Day / sequence	Global Day 110 of 365; calendar-free M04W4D5; optional date 2027-02-13.
2. Exact topic	Month 4 practical assessment and milestone review
3. Exact subtopics	Python, Bash, PowerShell, Git, regex, APIs, SQL, testing, logging and secure coding
4. Learning objective	Demonstrate intermediate scripting competence through an unfamiliar task
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 3-10 review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In 120 minutes, receive an unseen synthetic log format, write a parser, tests, normalization schema and short report without tutorial guidance. **2.** Standard (~2 h): In 120 minutes, receive an unseen synthetic log format, write a parser, tests, normalization schema and short report without tutorial guidance. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In 120 minutes, receive an unseen synthetic log format, write a parser, tests, normalization schema and short report without tutorial guidance. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M4-assessment.pdf and code repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement was ambiguous and how did you resolve it safely? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, independent problem-solving at least 11/15, and code runs cleanly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 3-10 review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D111 · M04W4D6 · Monthly review and prerequisite gate for security principles

Phase: Academic month · **Optional date:** 2027-02-14 · **Domain resources:** P01 / F09

1. Day / sequence	Global Day 111 of 365; calendar-free M04W4D6; optional date 2027-02-14.
2. Exact topic	Monthly review and prerequisite gate for security principles
3. Exact subtopics	Error-log analysis, debugging habits, test gaps, command notebook and spaced repetition
4. Learning objective	Close Month 4 gaps and verify readiness for core security theory
5. Primary lecture / reading + direct URL	Python for Everybody `https://www.py4e.com/` **Type:** University course · **Access:** Free site; Coursera specialization freemium
6. Exact section / lesson locator	Python for Everybody review chapters on files, dictionaries and regular expressions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework all failed assessment items, add missing tests, create ten flashcards, and map each Month 5 topic to a Month 1-4 prerequisite. **2.** Standard (~2 h): Rework all failed assessment items, add missing tests, create ten flashcards, and map each Month 5 topic to a Month 1-4 prerequisite. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework all failed assessment items, add missing tests, create ten flashcards, and map each Month 5 topic to a Month 1-4 prerequisite. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M4-review.md and prerequisite-map.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which programming weakness would block secure coding in Month 6? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and prerequisite map has no unexplained critical gap Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python for Everybody review chapters on files, dictionaries and regular expressions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D112 · M04W4D7 · Spaced revision, recovery and Month 5 preview

Phase: Academic month · **Optional date:** 2027-02-15 · **Domain resources:** P04 / F04 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 112 of 365; calendar-free M04W4D7; optional date 2027-02-15.
2. Exact topic	Spaced revision, recovery and Month 5 preview
3. Exact subtopics	Active recall of commands, APIs, regex, SQL and testing; portfolio cleanup; lighter workload
4. Learning objective	Preserve automation skill and prepare CIA/AAA/risk vocabulary
5. Primary lecture / reading + direct URL	RegexOne `https://regexone.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	RegexOne review lessons 10-15. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute mixed retrieval session, fix one README issue, review the CS50 Cybersecurity syllabus, and rest or take a light walkthrough of the Month 5 lab — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute mixed retrieval session, fix one README issue, review the CS50 Cybersecurity syllabus, and rest or take a light walkthrough of the Month 5 lab — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute mixed retrieval session, fix one README issue, review the CS50 Cybersecurity syllabus, and rest or take a light walkthrough of the Month 5 lab — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M4-consolidation.md and updated portfolio; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Explain how input validation, logging and least privilege appear in your own tool without notes Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All scheduled cards are reviewed and the three concepts are explained correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "RegexOne review lessons 10-15"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.